18

Superposition and Standing Waves

CHAPTER OUTLINE

18.1 Analysis Model: Waves in Interference

18.2 Standing Waves

18.3 Analysis Model: Waves Under Boundary Conditions

18.4 Resonance

18.5 Standing Waves in Air Columns

18.6 Standing Waves in Rods and Membranes

18.7 Beats: Interference in Time

18.8 Nonsinusoidal Wave Patterns

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ18.1 The ranking is (d) > (a) = (c) > (b). In the starting situation, the waves interfere constructively. When the sliding section is moved out by 0.1 m, the wave going through it has an extra path length of 0.2 m = $\lambda/4$, to show partial interference. When the slide has come out 0.2 m from the starting configuration, the extra path length is 0.4 m = $\lambda/2$, for destructive interference. Another 0.1 m and we are at $r_2 - r_1 = 3\lambda/4$ for partial interference as before. At last, another equal step of sliding and one wave travels one wavelength farther to interfere constructively.

OQ18.2 The fundamental frequency is described by

$$f_1 = \frac{v}{2L}, \text{ where } v = \left(\frac{T}{\mu}\right)^{1/2}$$

(i) Answer (e). If L is doubled, then the wavelength of the

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fundamental frequency is doubled, then $f = v/\lambda$ will be reduced by a factor of $\frac{1}{2}$.

(ii) Answer (d). If μ is doubled, then the speed is reduced by a factor of $\frac{1}{\sqrt{2}}$, so $f = v/\lambda$ will be reduced by a factor of $\frac{1}{\sqrt{2}}$.

(iii) Answer (b). If T is doubled, then the speed is increased by a factor of $\sqrt{2}$, so $f = v/\lambda$ will increase by a factor of $\sqrt{2}$.

OQ18.3 Answer (c). The two waves must have slightly different amplitudes at P because of their different distances, so they cannot cancel each other exactly.

OQ18.4 (i) Answer (e). If the end is fixed, there is inversion of the pulse upon reflection. Thus, when they meet, they cancel and the amplitude is zero.

(ii) Answer (c). If the end is free, there is no inversion on reflection. When they meet, the amplitude is $2A = 2(0.1\text{ m}) = 0.2\text{ m}$.

OQ18.5 Answer (a). At resonance, a tube closed at one end and open at the other forms a standing wave pattern with a node at the closed end and antinode at the open end. In the fundamental mode (or first harmonic), the length of the tube closed at one end is a quarter wavelength ($L = \lambda_1/4$ or $\lambda_1 = 4L$). Therefore, for the given tube, $\lambda_1 = 4(0.580\text{ m}) = 2.32\text{ m}$ and the fundamental frequency is

$$f_1 = \frac{v}{\lambda_1} = \frac{343\text{ m/s}}{2.32\text{ m}} = 148\text{ Hz}$$

OQ18.6 Answer (e). The number of beats per second (the beat frequency) equals the difference in the frequencies of the two tuning forks. Thus, if the beat frequency is 5 Hz and one fork is known to have a frequency of 245 Hz, the frequency of the second fork could be either $f_2 = 245\text{ Hz} - 5\text{ Hz} = 240\text{ Hz}$ or $f_2 = 245\text{ Hz} + 5\text{ Hz} = 250\text{ Hz}$. This means that the best answer for the question is choice (e), since choices (a) and (d) are both possibly correct.

OQ18.7 Answer (d). The tape will reduce the frequency of the fork, leaving the string frequency unchanged. If the bit of tape is small, the fork must have started with a frequency 4 Hz below that of the string, to end up with a frequency 5 Hz below that of the string. The string frequency is 262 + 4 = 266 Hz.

OQ18.8 Answer (c). The bow string is pulled away from equilibrium and released, similar to the way that a guitar string is pulled and released

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when it is plucked. Thus, standing waves will be excited in the bow string. If the arrow leaves from the exact center of the string, then a series of odd harmonics will be excited. Even harmonics will not be excited because they have a node at the point where the string exhibits its maximum displacement.

OQ18.9 Answer (d). The energy has not disappeared, but is still carried by the wave pulses. Each element of the string still has kinetic energy. This is similar to the motion of a simple pedulum. The pendulum does not stop at its equilibrium position during oscillation—likewise the elements of the string do not stop at the equilibrium position of the string when these two waves superimpose.

OQ18.10 Answer (c). On a string fixed at both ends, a standing wave with three nodes is the second harmonic: there is a node on each end and one in the middle, so it has two antinodes because there is an antinode between each pair of nodes. The number of antinodes is the same as the harmonic number. Doubling the frequency gives the fourth harmonic, therefore four antinodes.

OQ18.11 Answers (b) and (e). The strings have different linear densities and are stretched to different tensions, so they carry string waves with different speeds and vibrate with different fundamental frequencies. They are all equally long, so the string waves have equal fundamental wavelengths. They all radiate sound into air, where the sound moves with the same speed for different sound wavelengths.

OQ18.12 Answer (d). The resultant amplitude is greater than either individual amplitude, wherever the two waves are nearly enough in phase that $2A\cos(\phi/2)$ is greater than A. This condition is satisfied whenever the absolute value of the phase difference ϕ between the two waves is less than 120°.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ18.1 The resonant frequency depends on the length of the pipe. Thus, changing the length of the pipe will cause different frequencies to be emphasized in the resulting sound.

CQ18.2 No. The total energy of the pair of waves remains the same. Energy missing from zones of destructive interference appears in zones of constructive interference.

CQ18.3 What is needed is a tuning fork—or other pure-tone generator—of the desired frequency. Strike the tuning fork and pluck the corresponding string on the piano at the same time. If they are

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precisely in tune, you will hear a single pitch with no amplitude modulation. If the two frequences are a bit off, you will hear beats. As they vibrate, retune the piano string until the beat frequency goes to zero.

CQ18.4 Damping, and nonlinear effects in the vibration, transform the energy of vibration into internal energy.

CQ18.5 (a) The tuning fork hits the paper repetitively to make a sound like a buzzer, and the paper efficiently moves the surrounding air. The tuning fork will vibrate audibly for a shorter time.

(b) Instead of just radiating sound very softly into the surrounding air, the tuning fork makes the chalkboard vibrate. With its large area this stiff sounding board radiates sound into the air with higher power. So it drains away the fork's energy of vibration faster and the fork stops vibrating sooner.

(c) The tuning fork in resonance makes the column of air vibrate, especially at the antinode of displacement at the top of the tube. Its area is larger than that of the fork tines, so it radiates louder sound into the environment. The tuning fork will not vibrate for so long.

(d) The cardboard acts to cut off the path of air flow from the front to the back of a single tine. When a tine moves forward, the high pressure air in front of the tine can simply move to fill in the lower pressure area behind the tine. This "sloshing" of the air back and forth does not contribute to sound radiation and results in low intensity of sound actually leaving the tine. By cutting off this "sloshing" path by bringing the cardboard near, the tine becomes a more efficient radiator. This is the same theory as that involved with placing loudspeakers on baffles. A speaker enclosure for a loudspeaker is equivalent to an infinite baffle because there is no path the high pressure air can find to cancel the lower pressure air on the other side of the speaker.

CQ18.6 The loudness varies because of beats. The propellers are rotating at slightly different frequencies.

CQ18.7 Walking makes the person's hand vibrate a little. If the frequency of this motion is equal to the natural frequency of coffee sloshing from side to side in the cup, then a large-amplitude vibration of the coffee will build up in resonance. To get off resonance and back to the normal case of a small-amplitude disturbance producing a small-amplitude result, the person can walk faster, walk slower, or get a larger or smaller cup. You do not need a cover on your cup.

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CQ18.8 Consider the level of fluid in the bottle to be adjusted so that the air column above it resonates at the first harmonic. This is given by $f = \frac{v}{4L}$. This equation indicates that as the length L of the column increases (fluid level decreases), the resonant frequency decreases.

CQ18.9 No. Waves with all waveforms interfere. Waves with other wave shapes are also trains of disturbance that add together when waves from different sources move through the same medium at the same time.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 18.1 Analysis Model: Waves in Interference

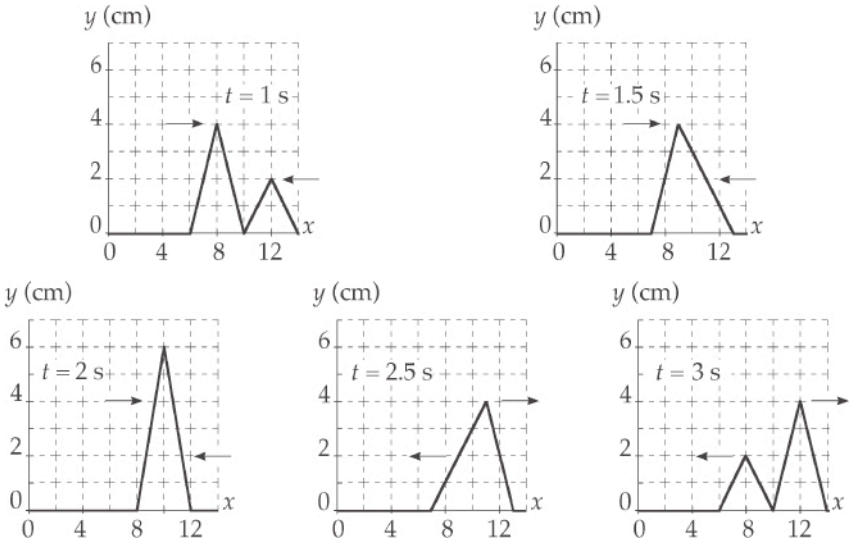
P18.1 Suppose the waves are sinusoidal. The sum is

$$(4.00\text{ cm})\sin(kx-\omega t)+(4.00\text{ cm})\sin(kx-\omega t+90.0°)$$
$$=2(4.00\text{ cm})\sin(kx-\omega t+45.0°)\cos 45.0°$$

So the amplitude of the resultant wave is

$$(8.00\text{ cm})\cos 45.0° = \boxed{5.66\text{ cm}}$$

P18.2 ANS. FIG. P18.2 shows the sketches at each of the times.



ANS. FIG. P18.2

P18.3 The superposition of the waves is given by

$$y = y_1 + y_2 = 3.00\cos(4.00x-1.60t)+4.00\sin(5.00x-2.00t)$$

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evaluated at the given x values.

(a) At $x = 1.00$, $t = 1.00$, the superposition of the two waves gives

$$y = 3.00\cos\left[4.00(1.00) - 1.60(1.00)\right] + 4.00\sin\left[5.00(1.00) - 2.00(1.00)\right]$$
$$= 3.00\cos(2.40\text{ rad}) + 4.00\sin(3.00\text{ rad}) = \boxed{-1.65\text{ cm}}$$

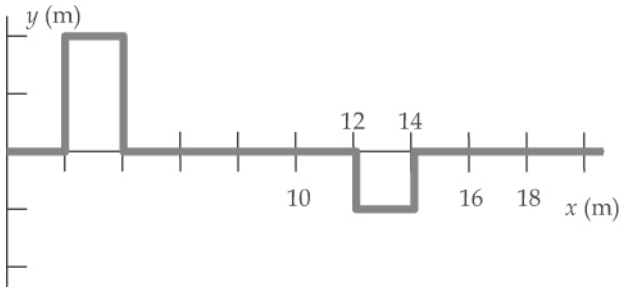
(b) At $x = 1.00$, $t = 0.500$, the superposition of the two waves gives

$$y = 3.00\cos\left[4.00(1.00) - 1.60(0.500)\right] + 4.00\sin\left[5.00(1.00) - 2.00(0.500)\right]$$
$$= 3.00\cos(3.20\text{ rad}) + 4.00\sin(4.00\text{ rad}) = \boxed{-6.02\text{ cm}}$$

(c) At $x = 0.500$, $t = 0$, the superposition of the two waves gives

$$y = 3.00\cos\left[4.00(1.00) - 1.60(0)\right] + 4.00\sin\left[5.00(1.00) - 2.00(0)\right]$$
$$= 3.00\cos(2.00\text{ rad}) + 4.00\sin(2.50\text{ rad}) = \boxed{+1.15\text{ cm}}$$

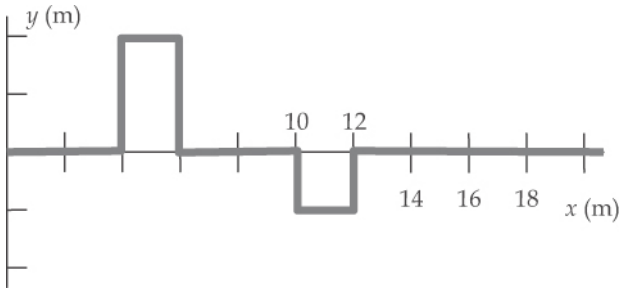
P18.4 (a) The graph at time $t = 0.00$ seconds is shown in ANS. FIG. P18.4(a)



ANS. FIG. P18.4(a)

The pulse initially on the left will move to the right at 1.00 m/s, and the one initially at the right will move toward the left at the same rate, as follows:

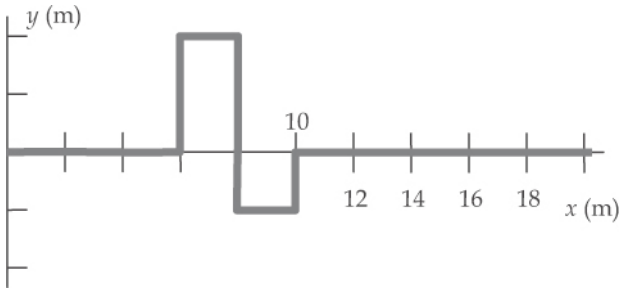
ANS. FIG. P18.4(b) shows the pulses at time t = 2.00 seconds



ANS. FIG. P18.4(b)

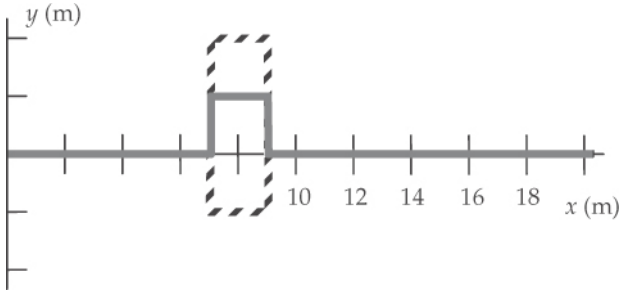
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ANS. FIG. P18.4(c) shows the waves at time t = 4.00 seconds, immediately before they overlap.



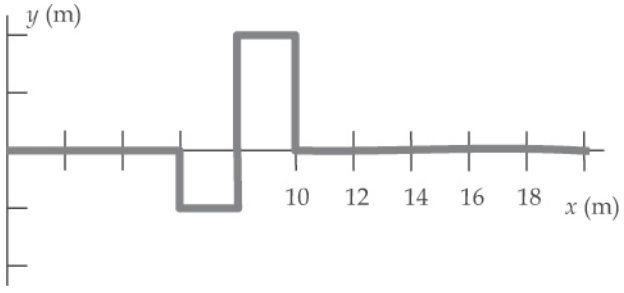
ANS. FIG. P18.4(c)

ANS. FIG. P18.4(d) shows the pulses at time t = 5.00 seconds, while the two pulses are fully overlapped. The two pulses are shown as dashed lines.



ANS. FIG. P18.4(d)

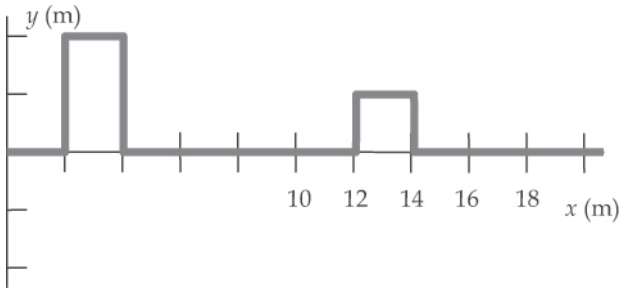
ANS. FIG. P18.4(e) shows the pulses at time At time t = 6.00 seconds, immediately after they completely pass.



ANS. FIG. P18.4(e)

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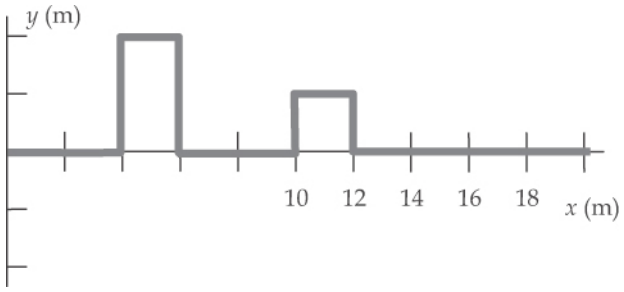
(b) If the pulse to the right is inverted, ANS. FIG. P18.4(f) shows the pulses at time t = 0.00 seconds.



ANS. FIG. P18.4(f)

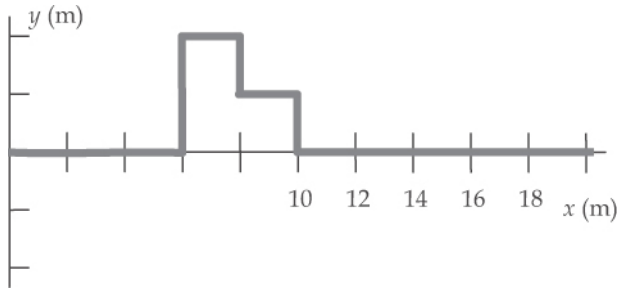
The pulse initially on the left will move to the right at 1.00 m/s, and the one initially at the right will move toward the left at the same rate, as follows:

ANS. FIG. P18.4(g) shows the two pulses at time t = 2.00 seconds



ANS. FIG. P18.4(g)

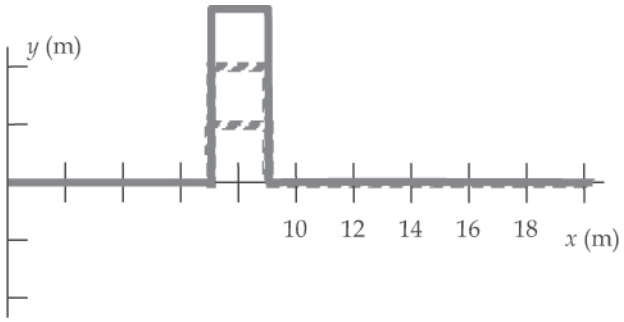
ANS. FIG. P18.4(h) shows the two pulses at time t = 4.00 seconds, immediately before they overlap.



ANS. FIG. P18.4(h)

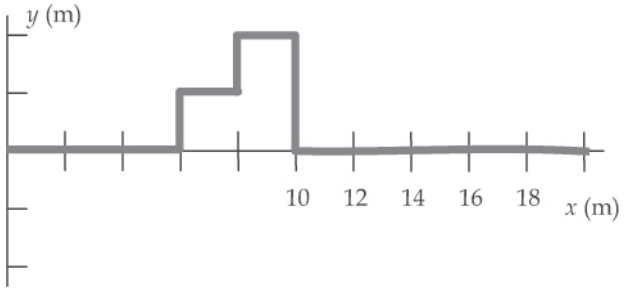
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ANS. FIG. P18.4(i) shows the two pulses at time t = 5.00 seconds, while the two pulses are fully overlapped. The two pulses are shown as dashed lines.



ANS. FIG. P18.4(i)

ANS. FIG. P18.4(j) shows the two pulses at time t = 6.00 seconds, immediately after they completely pass.



ANS. FIG. P18.4(j)

***P18.5** Waves reflecting from the near end travel 28.0 m (14.0 m down and 14.0 m back), while waves reflecting from the far end travel 66.0 m. The path difference for the two waves is:

$$\Delta r = 66.0 \text{ m} - 28.0 \text{ m} = 38.0 \text{ m}$$

Since $\lambda = \dfrac{v}{f}$,

Then

$$\frac{\Delta r}{\lambda} = \frac{(\Delta r) f}{v} = \frac{(38.0 \text{ m})(246 \text{ Hz})}{343 \text{ m/s}} = 27.254$$

or $\Delta r = 27.254\lambda$

The phase difference between the two reflected waves is then

$$\phi = (0.254)(1 \text{ cycle}) = (0.254)(2\pi \text{ rad}) = 1.594 \text{ rad} = \boxed{91.3°}$$

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P18.6 The wavelength of the sound emitted by the speaker is

$$\lambda = \frac{v}{f} = \frac{343 \text{ m/s}}{756 \text{ Hz}} \simeq 0.454 \text{ m}$$

Raising the sliding section by Δh changes the path through that section by $2\Delta h$, because sound must travel up and down through the addition distance.

(a) If constructive interference currently exists, this can be changed to destructive interference by increasing the path distance through the sliding section by $\lambda/2$, which means raising it by $\boxed{\lambda/4 = 0.113 \text{ m}}$.

(b) To move from constructive interference to the next occurrence of constructive interference, one should increase the path distance through the sliding section by λ, which means raising it by $\boxed{\lambda / 2 = 0.227 \text{ m}}$.

P18.7 (a) At constant phase, $\phi = 3x - 4t$ will be constant. Then $x = \dfrac{\phi + 4t}{3}$ will change: the wave moves. As t increases in this equation, x increases, so the first wave moves to the right, in the $\boxed{+x \text{ direction}}$.

In the same way, in the second case $x = \dfrac{\phi - 4t + 6}{3}$. As t increases, x must decrease, so the second wave moves to the left, in the $\boxed{-x \text{ direction}}$.

(b) We require that $y_1 + y_2 = 0$.

$$\frac{5}{(3x - 4t)^2 + 2} + \frac{-5}{(3x + 4t - 6)^2 + 2} = 0$$

This can be written as

$$(3x - 4t)^2 = (3x + 4t - 6)^2$$

Solving for the positive root, $8t = 6$, or

$\boxed{t = 0.750 \text{ s}}$

(c) The negative root yields

$$(3x - 4t) = -(3x + 4t - 6)$$

The time terms cancel, leaving $\boxed{x = 1.00 \text{ m}}$. At this point, the waves **always** cancel.

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P18.8 (a) $\Delta x = \sqrt{9.00\ \text{m}^2 + 4.00\ \text{m}^2} - 3.00\ \text{m} = \sqrt{13\ \text{m}^2} - 3.00\ \text{m} = 0.606\ \text{m}$

The wavelength is $\lambda = \dfrac{v}{f} = \dfrac{343\ \text{m/s}}{300\ \text{Hz}} = 1.14\ \text{m}.$

Thus, $\dfrac{\Delta x}{\lambda} = \dfrac{0.606}{1.14} = 0.530$ of a waves,

or $\Delta\phi = 2\pi(0.530) = \boxed{3.33\ \text{rad}}$.

(b) For destructive interference, we want

$$\frac{\Delta x}{\lambda} = 0.500 \rightarrow \lambda = \frac{\Delta x}{0.500} = 2\Delta x$$

The frequency is $f = \dfrac{v}{\lambda} = \dfrac{v}{2\Delta x} = \dfrac{343\ \text{m/s}}{2(0.606\ \text{m})} = \boxed{283\ \text{Hz}}$.

P18.9 The sum of two waves traveling in the same direction that have the same amplitude A_0, angular frequency ω, and wave number k but are different in phase ϕ have the resultant wave function in the form

$$y = 2A_0 \cos\left(\frac{\phi}{2}\right) \sin\left(kx - \omega t + \frac{\phi}{2}\right)$$

(a) $A = 2A_0 \cos\left(\dfrac{\phi}{2}\right) = 2(5.00\ \text{m}) \cos\left[\dfrac{-\pi/4}{2}\right] = \boxed{9.24\ \text{m}}$

(b) $f = \dfrac{\omega}{2\pi} = \dfrac{1\ 200\pi\ \text{rad/s}}{2\pi} = \boxed{600\ \text{Hz}}$

P18.10 Consider the geometry of the situation shown on the right. The path difference for the sound waves at the location of the man is

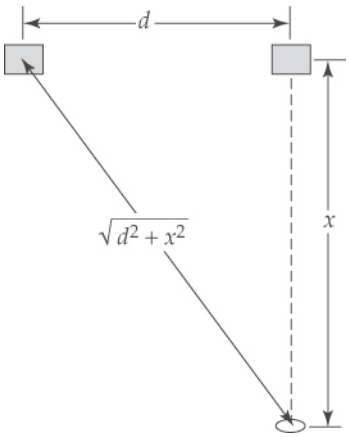
$$\Delta r = \sqrt{d^2 + x^2} - x$$

For a minimum, this path difference must equal a half-integral number of wavelengths:

$$\sqrt{d^2 + x^2} - x = \left(n + \tfrac{1}{2}\right)\lambda \qquad n = 0, 1, 2, \ldots$$

Solve for x:

$$x = \frac{d^2 - \left[\left(n + \frac{1}{2}\right)\lambda\right]^2}{2\left(n + \frac{1}{2}\right)\lambda}$$



ANS. FIG. P18.10

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In order for x to be positive, we must have

$$\left[\left(n+\frac{1}{2}\right)\lambda\right]^2 < d^2 \quad \rightarrow \quad n < \frac{d}{\lambda} - \frac{1}{2} = \frac{df}{v} - \frac{1}{2}$$

Substitute numerical values:

$$n < \frac{(4.00\text{ m})(200\text{ Hz})}{343\text{ m/s}} - \frac{1}{2} = 1.83$$

The only values of n that satisfy this requirement are $n = 0$ and $n = 1$.

Therefore,

the man walks through only *two* minima;
a third minimum is impossible

***P18.11** At any time and place, the phase shift between the waves is found by subtracting the phases of the two waves, $\Delta\phi = \phi_1 - \phi_2$.

$$\Delta\phi = (20.0\text{ rad/cm})x - (32.0\text{ rad/s})t$$
$$- [(25.0\text{ rad/cm})x - (40.0\text{ rad/s})t]$$

Collecting terms,

$$\Delta\phi = -(5.00\text{ rad/cm})x + (8.00\text{ rad/s})t$$

(a) At $x = 5.00$ cm and $t = 2.00$ s, the phase difference is

$$\Delta\phi = (-5.00\text{ rad/cm})(5.00\text{ cm}) + (8.00\text{ rad/s})(2.00\text{ s})$$

$$\Delta\phi = 9.00\text{ radians} = 516° = \boxed{156°}$$

(b) The sine functions repeat whenever their arguments change by an integer number of cycles, an integer multiple of 2π radians. Then the phase shift equals $\pm\pi$ whenever $\Delta\phi = \pi + 2n\pi$, for all integer values of n. Substituting this into the phase equation, we have

$$\pi + 2n\pi = -(5.00\text{ rad/cm})x + (8.00\text{ rad/s})t$$

At $t = 2.00$ s,

$$\pi + 2n\pi = -(5.00\text{ rad/cm})x + (8.00\text{ rad/s})(2.00\text{ s})$$

or $\quad (5.00\text{ rad/cm})x = (16.0 - \pi - 2n\pi)\text{ rad}$

The smallest positive value of x is found when $n = 2$:

$$x = \frac{(16.0 - 5\pi)\text{ rad}}{5.00\text{ rad/cm}} = \boxed{0.058\,4\text{ cm}}$$

***P18.12** $2A_0 \cos\left(\frac{\phi}{2}\right) = A_0$ so $\frac{\phi}{2} = \cos^{-1}\left(\frac{1}{2}\right) = 60.0° = \frac{\pi}{3}$

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Thus, the phase difference is

$$\phi = 120° = \frac{2\pi}{3}$$

This phase difference results if the time delay is

$$\frac{T}{3} = \frac{1}{3f} = \frac{\lambda}{3v}$$

$$\text{Time delay} = \frac{3.00 \text{ m}}{3(2.00 \text{ m/s})} = \boxed{0.500 \text{ s}}$$

P18.13 (a) First we calculate the wavelength: $\lambda = \frac{v}{f} = \frac{344 \text{ m/s}}{21.5 \text{ Hz}} = 16.0 \text{ m}$

Then we note that the path difference equals

$$9.00 \text{ m} - 1.00 \text{ m} = \frac{1}{2}\lambda$$

Point A is one-half wavelength farther from one speaker than from the other. The waves from the two sources interfere destructively, so the receiver records a minimum in sound intensity.

(b) We choose the origin at the midpoint between the speakers. If the receiver is located at point (x, y), then we must solve:

$$\sqrt{(x+5.00)^2 + y^2} - \sqrt{(x-5.00)^2 + y^2} = \frac{1}{2}\lambda$$

Then,

$$\sqrt{(x+5.00)^2 + y^2} = \sqrt{(x-5.00)^2 + y^2} + \frac{1}{2}\lambda$$

Square both sides and simplify to get

$$20.0x - \frac{\lambda^2}{4} = \lambda\sqrt{(x-5.00)^2 + y^2}$$

Upon squaring again, this reduces to

$$400x^2 - 10.0\lambda^2 x + \frac{\lambda^4}{16.0} = \lambda^2 (x-5.00)^2 + \lambda^2 y^2$$

Substituting, $\lambda = 16.0$ m, and reducing,

$$9.00x^2 - 16.0y^2 = 144$$

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Note that the equation $9.00x^2 - 16.0y^2 = 144$ represents two hyperbolas: one passes through the x axis at $x = +4.00$ m; the second, which is the mirror image of the first, passes through $x = -4.00$ m to the left of the y axis.

(c) Solve for y in terms of x:

$$9x^2 - 16y^2 = 144$$

Then

$$y = \pm\sqrt{\frac{9}{16}x^2 - 9} = \pm\frac{3}{4}x\sqrt{1 - \frac{16}{x^2}}$$

$$y = \pm\frac{3}{4}x\sqrt{1 - \frac{16}{x^2}}$$

For very large x, the square root term approaches 1:

$$y = \pm\frac{3}{4}x\sqrt{1 - \frac{16}{x^2}} \quad \rightarrow \quad y = \pm\frac{3}{4}x$$

To the right of the origin, for large x the hyperbola approaches the shape of a straight line above and below the x axis.

Yes; the limiting form of the path is two straight lines through the origin with slope ±0.75.

Section 18.2 Standing Waves

P18.14 (a) From the resultant wave $y = 2A\sin\left(kx + \frac{\phi}{2}\right)\cos\left(\omega t - \frac{\phi}{2}\right)$,

the shape of the wave form is determined by the term

$$\sin\left(kx + \frac{\phi}{2}\right).$$

The nodes are located at $kx + \frac{\phi}{2} = n\pi$, or where $x = \frac{n\pi}{k} - \frac{\phi}{2k}$.

The separation of adjacent nodes is

$$\Delta x = \left[(n+1)\frac{\pi}{k} - \frac{\phi}{2k}\right] - \left[\frac{n\pi}{k} - \frac{\phi}{2k}\right] = \frac{\pi}{k} = \frac{\lambda}{2}$$

The nodes are still separated by half a wavelength.

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(b) Yes. The nodes are located at $kx + \frac{\phi}{2} = n\pi$, so that $x = \frac{n\pi}{k} - \frac{\phi}{2k}$, which means that each node is shifted $\frac{\phi}{2r}$ to the left by the phase difference between the traveling waves in comparison to the case in which $\phi = 0$.

P18.15 $y = (1.50\text{ m}) \sin(0.400x) \cos(200t) = 2A_0 \sin kx \cos \omega t$

Compare corresponding parts:

(a) $k = \frac{2\pi}{\lambda} = 0.400\text{ rad/m}$

$$\lambda = \frac{2\pi}{0.400\text{ rad/m}} = \boxed{15.7\text{ m}}$$

(b) $\omega = 2\pi f$ so $f = \frac{\omega}{2\pi} = \frac{200\text{ rad/s}}{2\pi\text{ rad}} = \boxed{31.8\text{ Hz}}$

(c) The speed of waves in the medium is

$$v = \lambda f = \frac{\lambda}{2\pi} 2\pi kf = \frac{\omega}{k} = \frac{200\text{ rad/s}}{0.400\text{ rad/m}} = \boxed{500\text{ m/s}}$$

P18.16 From $y = 2A_0 \sin kx \cos \omega t$, we find

$$\frac{\partial y}{\partial x} = 2A_0 k \cos kx \cos \omega t \qquad \frac{\partial y}{\partial t} = -2A_0 \omega \sin kx \sin \omega t$$

$$\frac{\partial^2 y}{\partial x^2} = -2A_0 k^2 \sin kx \cos \omega t$$

$$\frac{\partial^2 y}{\partial t^2} = -2A_0 \omega^2 \sin kx \cos \omega t$$

Substitution into the wave equation gives

$$-2A_0 k^2 \sin kx \cos \omega t = \left(\frac{1}{v^2}\right)\left(-2A_0 \omega^2 \sin kx \cos \omega t\right)$$

This is satisfied, provided that $v = \frac{\omega}{k}$. But this is true, because

$$v = \lambda f = \frac{\lambda}{2\pi} 2\pi f = \frac{\omega}{k}$$

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P18.17 $y_1 = 3.00 \sin\left[\pi(x + 0.600t)\right]$; $y_2 = 3.00 \sin\left[\pi(x - 0.600t)\right]$

$$y = y_1 + y_2 = \left[3.00 \sin(\pi x)\cos(0.600\pi t) + 3.00 \sin(\pi x)\cos(0.600\pi t)\right]$$

$$y = (6.00 \text{ cm}) \sin(\pi x)\cos(0.600\pi t)$$

We can take $\left|\cos(0.600\pi t)\right| = 1$ to get the maximum y.

(a) At $x = 0.250$ cm, $\left|y_{max}\right| = \left|(6.00 \text{ cm}) \sin(0.250\pi)\right| = \boxed{4.24 \text{ cm}}$

(b) At $x = 0.500$ cm, $\left|y_{max}\right| = \left|(6.00 \text{ cm}) \sin(0.500\pi)\right| = \boxed{6.00 \text{ cm}}$

(c) At $x = 1.50$ cm, $\left|y_{max}\right| = \left|(6.00 \text{ cm}) \sin(1.50\pi)\right| = \boxed{6.00 \text{ cm}}$

(d) The antinodes occur where

$$\sin(\pi x) = \pm 1 \quad \rightarrow \quad \pi x = n\frac{\pi}{2}$$

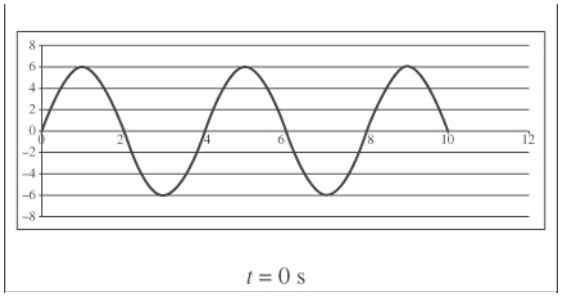
or where $x = \frac{n}{2}$, where $n = 1,\ 3,\ 5,\ 7, \ldots$ and x is in centimeters.

$n = 1$: $x_1 = \frac{1}{2} = \boxed{0.500 \text{ cm}}$ as in (b)

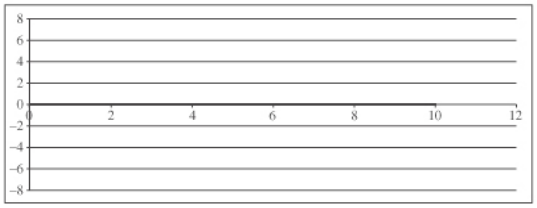
$n = 3$: $x_2 = \frac{3}{2} = \boxed{1.50 \text{ cm}}$ as in (c)

$n = 5$: $x_3 = \frac{5}{2} = \boxed{2.50 \text{ cm}}$

P18.18 (a) ANS. FIG. P18.18 shows the graphs for $t = 0$, $t = 5$ ms, $t = 10$ ms, $t = 15$ ms, and $t = 20$ ms. The units of the x and y axes are meters.

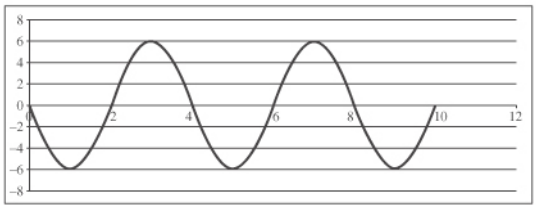


$t = 0$ s

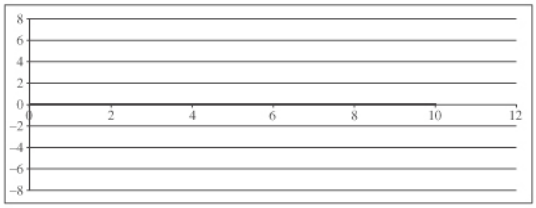


$t = 5$ ms

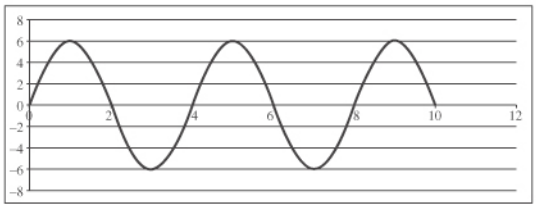
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$t = 10$ ms



$t = 15$ ms



$t = 20$ ms

ANS. FIG. P18.18

(b) In any one picture, the wavelength is the smallest distance along the x axis that contains a nonrepeating shape. The wavelength is $\lambda = 4$ m.

(c) The frequency is the inverse of the period. The period is the time the wave takes to go from a full amplitude starting shape to the inversion of that shape and then back to the original shape. The period is the time interval between the top and bottom graphs: 20 ms. The frequency is $1/0.020$ s $= 50$ Hz.

(d) 4 m. By comparison with the wave function $y = (2A \sin kx) \cos \omega t$, we identify $k = \pi/2$, and then compute $\lambda = 2\pi/k$.

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(e) 50 Hz. By comparison with the wave function $y = (2A \sin kx)\cos \omega t$, we identify $\omega = 2\pi f = 100\pi$.

P18.19 The facing speakers produce a standing wave in the space between them, with the spacing between nodes being

$$d_{NN} = \frac{\lambda}{2} = \frac{v}{2f} = \frac{343 \text{ m/s}}{2\left(800 \text{ s}^{-1}\right)} = 0.214 \text{ m}$$

If the speakers vibrate in phase, the point halfway between them is an antinode of pressure at a distance from either speaker of

$$\frac{1.25 \text{ m}}{2} = 0.625 \text{ m}$$

Then there is a node one-quarter of a wavelength away at

$$0.625 - \frac{0.214}{2} = \boxed{0.518 \text{ m}}$$

from either speaker, after which, there is a node every half-wavelength:

a node at $0.518 \text{ m} - 0.214 \text{ m} = \boxed{0.303 \text{ m}}$

a node at $0.303 \text{ m} - 0.214 \text{ m} = \boxed{0.089\ 1 \text{ m}}$

a node at $0.518 \text{ m} + 0.214 \text{ m} = \boxed{0.732 \text{ m}}$

a node at $0.732 \text{ m} + 0.214 \text{ m} = \boxed{0.947 \text{ m}}$

and a node at $0.947 \text{ m} + 0.214 \text{ m} = \boxed{1.16 \text{ m}}$ from either speaker.

Section 18.3 Analysis Model: Waves Under Boundary Conditions

***P18.20** We are given $L = 120$ cm, $f = 120$ Hz.

(a) For four segments, $L = 2\lambda$ or $\lambda = 60.0 \text{ cm} = \boxed{0.600 \text{ m}}$.

(b) $v = \lambda f = 72.0 \text{ m/s}$, $f_1 = \frac{v}{2L} = \frac{72.0 \text{ m/s}}{2(1.20 \text{ m})} = \boxed{30.0 \text{ Hz}}$

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P18.21 Using L_v for the vibrating portion of the string of total length L,

$$f = \frac{v}{2L_v} = \frac{1}{2L_v}\sqrt{\frac{T}{\mu}} = \frac{1}{2L_v}\sqrt{\frac{MgL}{m}}$$

$$= \frac{1}{2(4.00\text{ m})}\sqrt{\frac{(4.00\text{ kg})(9.80\text{ m/s}^2)(5.00\text{ m})}{0.008\,00\text{ kg}}} = \boxed{19.6\text{ Hz}}$$

P18.22 The frequency of vibration of a string is determined by the wave speed and the wavelength of the standing wave on the string. The length of the string and mode number n determines the size of the allowed wavelengths:

$$\lambda = 2L/n$$

$$f = \frac{v}{\lambda} = \frac{v}{2L/n} = n\frac{v}{2L}$$

As long as the wave speed does not change,

$$f \propto \frac{n}{L}$$

and so we may compare frequencies of vibrations for different modes and lengths of string:

$$\frac{f_2}{f_1} = \frac{n_2 L_1}{n_1 L_2}$$

When the string is pressed down on the fret, the wave speed on the string remains the same, but the length of the vibrating string is smaller. When the string is plucked, it vibrates at the fundamental frequency ($n = 1$) corresponding to the shorter length of the string. We can compare frequencies and length of vibrating string thus:

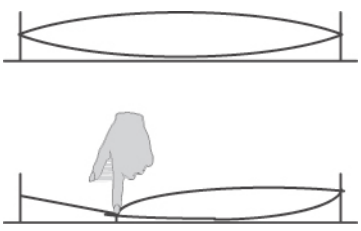
$$\frac{f_2}{f_1} = \frac{n_2 L_1}{n_1 L_2}$$

For the original length of string, $L_1 = L = 0.640$ m, $n_1 = 1$, and $f_1 = 330$ Hz.

(a) When the string is stopped at the fret, $L_2 = \frac{2}{3}L_1$, and $n_1 = n_2 = 1$.

$$\frac{f_2}{f_1} = \frac{n_2 L_1}{n_1 L_2} = \frac{(1)L_1}{(1)\left(\frac{2}{3}L_1\right)} = \frac{3}{2}$$

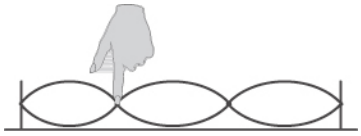
$$f_2 = \frac{3}{2}f_1 = \boxed{495\text{ Hz}}$$



ANS. FIG. P18.22(a)

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(b) The light touch at a point one-third of the way along the string forces the point of contact to be a node while still allowing the entire string to vibrate. The whole string vibrates in three loops; therefore, the string vibrates in its third resonance possibility ($n = 3$):

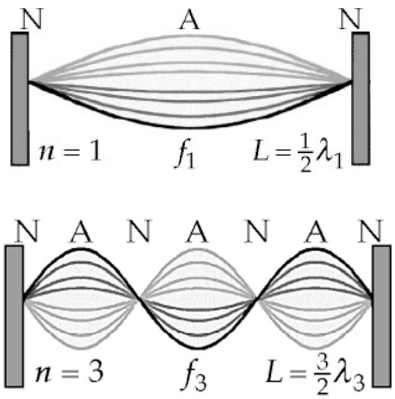


ANS. FIG. P18.22(b)

$$\frac{f_2}{f_1} = \frac{n_2 L_1}{n_1 L_2}$$

$$\frac{f_2}{f_1} = \frac{(3)L_1}{(1)L_1} \rightarrow f_2 = 3f_1 = \boxed{990 \text{ Hz}}$$

P18.23 When the string vibrates in the lowest frequency mode, the length of string forms a standing wave where $L = \lambda/2$, so the fundamental harmonic wavelength is



ANS. FIG. P18.23

$$\lambda = 2L = 2(0.700 \text{ m})$$
$$= 1.40 \text{ m}$$

and the speed is

$$v = \lambda f = \left(220 \text{ s}^{-1}\right)(1.40 \text{ m})$$
$$= 308 \text{ m/s}$$

(a) From the tension equation

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{m/L}}$$

we get $T = v^2 m/L$,

or $$T = \frac{(308 \text{ m/s})^2 \left(1.20 \times 10^{-3} \text{ kg}\right)}{0.700 \text{ m}} = \boxed{163 \text{ N}}$$

(b) For the third harmonic, the tension, linear density, and speed are the same, but the string vibrates in three segments. Thus, the wavelength is one third as long as in the fundamental.

$$\lambda_3 = \lambda_1/3$$

From the equation $v = f\lambda$, we find the frequency is three times as high.

$$f_3 = \frac{v}{\lambda_3} = 3\frac{v}{\lambda_1} = 3f_1 = \boxed{660 \text{ Hz}}$$

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P18.24 (a) Because the string is taut and is fixed at both ends, any standing waves will have nodes (which are multiples of $\lambda/2$ apart). The wavelengths of all possible modes on the string are:

$$\lambda_n = \frac{2L}{n}, \text{ where } n = 1, 2, 3,\ldots$$

The fundamental ($n = 1$) wavelength must then have a wavelength λ exactly twice the string length, or

$$\lambda_1 = \frac{2L}{1} = 2(2.60 \text{ m}) = \boxed{5.20 \text{ m}}$$

(b) $\boxed{\text{No. We do not know the speed of waves on the string.}}$ To obtain the frequencies on the string,

$$f_n = n\frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

it is necessary to have either the wave velocity v or the tension T and mass density μ of the string. We do not know these; therefore, it is not possible to find the frequency of this mode on the string.

P18.25 Because the piano string is fixed at both end, it will have nodes at each end, and also a node between the two antinodes. Thus, this standing wave pattern represents one full wavelength.

(a) Thus, this is $\boxed{\text{second harmonic}}$.

(b) And, because $\lambda_n = \frac{2L}{n}$, where $n = 1, 2, 3,\ldots$

The wavelength is $\lambda_2 = \frac{2L}{n} = \frac{2(74.0 \text{ cm})}{2} = \boxed{74.0 \text{ cm}}$.

(c) Because nodes are at both ends and in the middle, the number of nodes is $\boxed{3}$.

P18.26 The wave speed is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{20.0 \text{ N}}{9.00\times10^{-3} \text{ kg/m}}} = 47.1 \text{ m/s}$$

For a vibrating string of length L fixed at both ends, there are nodes at both ends. The wavelength of the fundamental is $\lambda = 2d_{NN} = 2L = 0.600$ m, and the frequency is

$$f_1 = \frac{v}{\lambda} = \frac{v}{2L} = \frac{47.1 \text{ m/s}}{0.600 \text{ m}} = \boxed{78.6 \text{ Hz}}$$

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After NAN, the next three vibration possibilities read NANAN, NANANAN, and NANANANAN. Each has just one more node and one more antinode than the one before. Respectively, these string waves have wavelengths of one-half, one-third, and one-quarter of 60.0 cm. The harmonic frequencies are

$$f_2 = 2f_1 = \boxed{157\ \text{Hz}}$$

$$f_3 = 3f_1 = \boxed{236\ \text{Hz}}$$

$$f_4 = 4f_1 = \boxed{314\ \text{Hz}}$$

P18.27 (a) Let n be the number of nodes in the standing wave resulting from the 25.0-kg mass. Then $n + 1$ is the number of nodes for the standing wave resulting from the 16.0-kg mass. For standing waves, $\lambda = \dfrac{2L}{n}$ and the frequency is $f = \dfrac{v}{\lambda}$. The frequency does not change as the masses are changed.

Thus, $\quad f = \dfrac{n}{2L}\sqrt{\dfrac{T_n}{\mu}} \quad$ and also $\quad f = \dfrac{n+1}{2L}\sqrt{\dfrac{T_{n+1}}{\mu}}.$

Equating the expressions for f, we have

$$\frac{n+1}{n} = \sqrt{\frac{T_n}{T_{n+1}}} = \sqrt{\frac{(25.0\ \text{kg})g}{(16.0\ \text{kg})g}} = \frac{5}{4}$$

Therefore, $4n + 4 = 5n$, or $n = 4$. Using either expression for f, we find

$$f = \frac{4}{2(2.00\ \text{m})}\sqrt{\frac{(25.0\ \text{kg})(9.80\ \text{m/s}^2)}{0.002\,00\ \text{kg/m}}} = \boxed{350\ \text{Hz}}$$

(b) For tension $T_n = mg$, we write

$$f = \frac{n}{2L}\sqrt{\frac{T_n}{\mu}} = \frac{n}{2L}\sqrt{\frac{mg}{\mu}} \quad \rightarrow \quad m = \frac{4L^2 f^2 \mu}{n^2 g}$$

We solve for m for $n = 1$:

$$m = \frac{4(2.00\ \text{m})^2(350\ \text{Hz})^2(0.002\,00\ \text{kg/m})}{(1)^2(9.80\ \text{m/s}^2)} = \boxed{400\ \text{kg}}$$

***P18.28** (a) For a standing wave of 6 loops, $6(\lambda/2) = L$, or

$$\lambda = L/3 = (2.00\ \text{m})/3$$

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The speed of the waves in the string is then

$$v = \lambda f = \left(\frac{2.00\text{ m}}{3}\right)\left(150\text{ Hz}^{-1}\right) = 1.00\times10^{2}\text{ m/s}$$

Since the tension in the string is

$$F = mg = \left(5.00\text{ kg}\right)\left(9.80\text{ m/s}^{2}\right) = 49.0\text{ N}$$

$v = \sqrt{\dfrac{F}{\mu}}$ gives

$$\mu = \frac{F}{v^{2}} = \frac{49.0\text{ N}}{\left(1.00\times10^{2}\text{ m/s}\right)^{2}} = \boxed{4.90\times10^{-3}\text{ kg/m}}$$

(b) If m = 45.0 kg, then

$$F = mg = \left(45.0\text{ kg}\right)\left(9.80\text{ m/s}^{2}\right) = 4.41\times10^{2}\text{ N}$$

and

$$v = \sqrt{\frac{4.41\times10^{2}\text{ N}}{4.90\times10^{-3}\text{ kg/m}}} = 3.00\times10^{2}\text{ m/s}$$

Thus, the wavelength will be

$$\lambda = \frac{v}{f} = \frac{3.00\times10^{2}\text{ m/s}}{150\text{ Hz}} = 2.00\text{ m}$$

and the number of loops is

$$n = \frac{L}{\lambda/2} = \frac{2.00\text{ m}}{1.00\text{ m}} = \boxed{2}$$

(c) If m = 10.0 kg, the tension is

$$F = mg = \left(10.0\text{ kg}\right)\left(9.80\text{ m/s}^{2}\right) = 98.0\text{ N}$$

and

$$v = \sqrt{\frac{98.0\text{ N}}{4.90\times10^{-3}\text{ kg/m}}} = 1.41\times10^{2}\text{ m/s}$$

Then,

$$\lambda = \frac{v}{f} = \frac{1.41\times10^{2}\text{ m/s}}{150\text{ Hz}} = 0.943\text{ m}$$

and $n = \dfrac{L}{\lambda/2} = \dfrac{2.00\text{ m}}{0.471\text{ m}}$ is not an integer,

so $\boxed{\text{no standing wave will form}}$.

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P18.29 In the fundamental mode, the string above the rod has only two nodes, at A and B, with an antinode halfway between A and B. Thus,

$$\frac{\lambda}{2} = \overline{AB} = \frac{L}{\cos\theta} \quad \text{or} \quad \lambda = \frac{2L}{\cos\theta}$$

Since the fundamental frequency is f, the wave speed in this segment of string is

$$v = \lambda f = \frac{2Lf}{\cos\theta}$$

Because of the pulley, the string has tension $T = Mg$.

Also,

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg}{m/\overline{AB}}} = \sqrt{\frac{MgL}{m\cos\theta}}$$

Thus,

$$\frac{2Lf}{\cos\theta} = \sqrt{\frac{MgL}{m\cos\theta}} \quad \text{or} \quad \frac{4L^2f^2}{\cos^2\theta} = \frac{MgL}{m\cos\theta}$$

and the mass of string above the rod is:

$$m = \frac{Mg\cos\theta}{4f^2L} = \frac{(1.00\text{ kg})(9.80\text{ m/s}^2)\cos 35.0°}{4(60.0\text{ Hz})^2(0.300\text{ m})} = \boxed{1.86\text{ g}}$$

P18.30 In the fundamental mode, the string above the rod has only two nodes, at A and B, with an anti-node halfway between A and B. Thus,

$$\frac{\lambda}{2} = \overline{AB} = \frac{L}{\cos\theta} \quad \text{or} \quad \lambda = \frac{2L}{\cos\theta}$$

Since the fundamental frequency is f, the wave speed in this segment of string is

$$v = \lambda f = \frac{2Lf}{\cos\theta}$$

Because of the pulley, the string has tension $T = Mg$.

Also,

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg}{m/\overline{AB}}} = \sqrt{\frac{MgL}{m\cos\theta}}$$

Thus,

$$\frac{2Lf}{\cos\theta} = \sqrt{\frac{MgL}{m\cos\theta}} \quad \text{or} \quad \frac{4L^2f^2}{\cos^2\theta} = \frac{MgL}{m\cos\theta}$$

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and the mass of string above the rod is:

$$\boxed{m = \frac{Mg\cos\theta}{4f^2L}}$$

P18.31 When the open string vibrates in its fundamental mode it produces concert G. When concert A is played, the shorter length of string vibrates in its fundamental mode also.

(a) $\lambda_G = 2L_G = \dfrac{v}{f_G};\ \lambda_A = 2L_A = \dfrac{v}{f_A},$ and $\dfrac{L_A}{L_G} = \dfrac{f_G}{f_A}$

$$L_G - L_A = L_G - \left(\frac{L_A}{L_G}\right)L_G = L_G - \left(\frac{f_G}{f_A}\right)L_G = L_G\left(1 - \frac{f_G}{f_A}\right)$$

$$L_G - L_A = (0.350\ \text{m})\left(1 - \frac{392}{440}\right) = 0.038\ 2\ \text{m}$$

Thus, $L_A = L_G - 0.038\ 2\ \text{m} = 0.350\ \text{m} - 0.038\ 2\ \text{m} = 0.312\ \text{m},$

or the finger should be placed $\boxed{31.2\ \text{cm from the bridge}}$.

(b) If the position of the finger is correct within $dL = 0.600$ cm when the note is played, by how much can the tension be off so that the note is the same? We want to find the maximum allowable percentage change in tension, dT/T, that will compensate for a small percentage change in position, dL/L, so that the change in the fundamental frequency, df, is zero.

From the expression for the fundamental frequency,

$$f = \frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}, \quad \text{we require } df = 0.$$

$$df = \frac{-dL}{2L^2}\sqrt{\frac{T}{\mu}} + \frac{1}{2L}\frac{1}{2}\frac{dT}{\sqrt{T\mu}} = 0 \quad \rightarrow \quad \frac{dL}{2L^2}\sqrt{\frac{T}{\mu}} = \frac{1}{4L}\frac{dT}{\sqrt{T\mu}}$$

$$\rightarrow \quad \frac{dL}{L}\sqrt{\frac{T}{\mu}} = \frac{1}{2}\sqrt{\frac{T}{\mu}}\frac{dT}{T} \quad \rightarrow \quad \frac{dT}{T} = 2\frac{dL}{L}$$

$$= 2\left(\frac{0.600\ \text{cm}}{31.2\ \text{cm}}\right) \quad \rightarrow \quad \boxed{3.85\%}$$

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P18.32 Let $m = \rho V$ represent the mass of the copper cylinder. The original tension in the wire is $T_1 = mg = \rho Vg$. The water exerts a buoyant force $\rho_{\text{water}}\left(\frac{V}{2}\right)g$ on the cylinder, to reduce the tension to

$$T_2 = \rho Vg - \rho_{\text{water}}\left(\frac{V}{2}\right)g = \left(\rho - \frac{\rho_{\text{water}}}{2}\right)Vg$$

The speed of a wave on the string changes from $\sqrt{\frac{T_1}{\mu}}$ to $\sqrt{\frac{T_2}{\mu}}$. The frequency changes from

$$f_1 = \frac{v_1}{\lambda} = \sqrt{\frac{T_1}{\mu}}\frac{1}{\lambda} \quad \text{to} \quad f_2 = \sqrt{\frac{T_2}{\mu}}\frac{1}{\lambda}$$

where we assume $\lambda = 2L$ is constant.

Then

$$\frac{f_2}{f_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{\rho - \rho_{\text{water}}/2}{\rho}} = \sqrt{\frac{8.92 - 1.00/2}{8.92}}$$

and

$$f_2 = (300\text{ Hz})\sqrt{\frac{8.42}{8.92}} = \boxed{291\text{ Hz}}$$

P18.33 Comparing $y = 0.002\,00 \sin(\pi x)\cos(100\pi t)$ with $y = 2A \sin kx \cos \omega t$,

we find $k = \frac{2\pi}{\lambda} = \pi\text{ m}^{-1} \rightarrow \lambda = 2.00\text{ m}$, and

$\omega = 2\pi f = 100\pi\text{ s}^{-1} \rightarrow f = 50.0\text{ Hz}$

(a) The distance between adjacent nodes is $d_{\text{NN}} = \frac{\lambda}{2} = 1.00\text{ m}$,

and on the string there are $\frac{L}{d_{\text{NN}}} = \frac{3.00\text{ m}}{1.00\text{ m}} = \boxed{3\text{ loops}}$.

(b) For the speed we have $v = \omega/k = 100\pi\text{ s}^{-1}/\pi\text{ m}^{-1} = 100\text{ m/s}$.

In the simplest standing wave vibration, $d_{\text{NN}} = 3.00\text{ m} = \frac{\lambda_b}{2}$,

$\lambda_b = 6.00\text{ m}$, and $f_b = \frac{v_a}{\lambda_b} = \frac{100\text{ m/s}}{6.00\text{ m}} = \boxed{16.7\text{ Hz}}$.

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(c) In $v_0 = \sqrt{\frac{T_0}{\mu}}$, if the tension increases to $T_c = 9T_0$ and the string does not stretch, the speed increases to

$$v_c = \sqrt{\frac{9T_0}{\mu}} = 3\sqrt{\frac{T_0}{\mu}} = 3v_0 = 3(100\ \text{m/s}) = 300\ \text{m/s}$$

Then $\lambda_c = \frac{v_c}{f_a} = \frac{300\ \text{m/s}}{50\ \text{Hz}} = 6.00\ \text{m}, \quad d_{\text{NN}} = \frac{\lambda_c}{2} = 3.00\ \text{m},$

and $\boxed{\text{one loop}}$ fits onto the string.

Section 18.4 Resonance

P18.34 The wave speed is $v = \sqrt{gd} = \sqrt{(9.80\ \text{m/s}^2)(36.1\ \text{m})} = 18.8\ \text{m/s}.$

The bay has one end open and one closed. Its simplest resonance is with a node of horizontal velocity, which is also an antinode of vertical displacement, at the head of the bay and an antinode of velocity, which is a node of displacement, at the mouth.

Then, $d_{\text{NA}} = 210 \times 10^3\ \text{m} = \frac{\lambda}{4}$

and $\lambda = 840 \times 10^3\ \text{m}.$

Therefore, the period is

$$T = \frac{1}{f} = \frac{\lambda}{v} = \frac{840 \times 10^3\ \text{m}}{18.8\ \text{m/s}} = 4.47 \times 10^4\ \text{s} = \boxed{12\ \text{h, 24 min}}$$

The natural frequency of the water sloshing in the bay agrees precisely with that of lunar excitation, so we identify the extra-high tides as amplified by resonance.

P18.35 (a) The wave speed is $v = \frac{9.15\ \text{m}}{2.50\ \text{s}} = \boxed{3.66\ \text{m/s}}$.

(b) There are antinodes at both ends of the pond, so the distance between adjacent antinodes is $d_{\text{AA}} = \frac{\lambda}{2} = 9.15\ \text{m}$ and the wavelength is $\lambda = 18.3\ \text{m}$

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The frequency is then $f = \frac{v}{\lambda} = \frac{3.66 \text{ m/s}}{18.3 \text{ m}} = \boxed{0.200 \text{ Hz}}$.

We have assumed the wave speed on water is the same for all wavelengths.

P18.36 The distance between adjacent nodes is one-quarter of the circumference.

$$d_{NN} = d_{AA} = \frac{\lambda}{2} = \frac{20.0 \text{ cm}}{4} = 5.00 \text{ cm}$$

so $\lambda = 10.0 \text{ cm}$

and $f = \frac{v}{\lambda} = \frac{900 \text{ m/s}}{0.100 \text{ m}} = 9\,000 \text{ Hz} = \boxed{9.00 \text{ kHz}}$

The singer must match this frequency quite precisely for some interval of time to feed enough energy into the glass to crack it.

Section 18.5 Standing Waves in Air Columns

***P18.37** Assuming an air temperature of $T = 37.0°\text{C} = 310 \text{ K}$, the speed of sound inside the pipe is

$$v = 331 \text{ m/s} + (0.600 \text{ m/s} \cdot °\text{C})(37.0°\text{C}) = 353 \text{ m/s}$$

In the fundamental resonant mode, the wavelength of sound waves in a pipe closed at one end is $\lambda = 4L$. Thus, for the whooping crane,

$$\lambda = 4(5.00 \text{ ft}) = 2.00 \times 10^1 \text{ ft}$$

and

$$f = \frac{v}{\lambda} = \frac{(353 \text{ m/s})}{(2.00 \times 10^1 \text{ ft})}\left(\frac{3.281 \text{ ft}}{1 \text{ m}}\right) = \boxed{57.9 \text{ Hz}}$$

18.38 At $T = 37.0°\text{C} = 310 \text{ K}$, the speed of sound in air is

$$v = (331 \text{ m/s})\sqrt{1 + \frac{T_C}{273}} = (331 \text{ m/s})\sqrt{1 + \frac{37.0}{273}} = 353 \text{ m/s}$$

Thus, the wavelength of 3 000-Hz sound is

$$\lambda = \frac{v}{f} = \frac{353 \text{ m/s}}{3\,000 \text{ Hz}} = 0.118 \text{ m}$$

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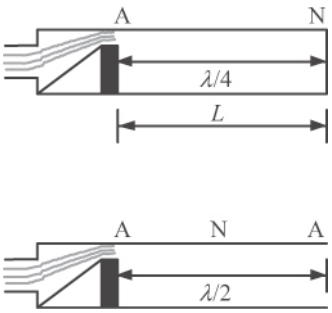
For the fundamental resonant mode in a pipe closed at one end, the length required is

$$L = \frac{\lambda}{4} = \frac{0.118 \text{ m}}{4} = 0.029\ 4 \text{ m} = \boxed{2.94 \text{ cm}}$$

P18.39 (a) For the fundamental mode in a closed pipe, $\lambda = 4L$, as in the diagram.

But $v = f\lambda$, therefore $L = \dfrac{v}{4f}$.

so, $L = \dfrac{343 \text{ m/s}}{4\left(240 \text{ s}^{-1}\right)} = \boxed{0.357 \text{ m}}$



ANS. FIG. P18.39

(b) For an open pipe, $\lambda = 2L$, as in the diagram.

So, $L = \dfrac{v}{2f} = \dfrac{343 \text{ m/s}}{2\left(240 \text{ s}^{-1}\right)} = \boxed{0.715 \text{ m}}$

P18.40 The 32.0-cm length corresponds to $d_{AA} = 0.320 \text{ m}$, which gives a wavelength of

$$\lambda = 2d_{AA} = 2(0.320 \text{ m}) = 0.640 \text{ m}$$

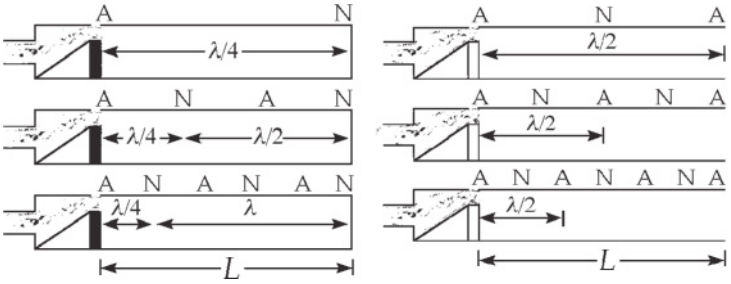
(a) The frequency of the lowest note is

$$f = \frac{v}{\lambda} = \frac{343 \text{ m/s}}{0.640 \text{ m}} = \boxed{536 \text{ Hz}}$$

(b) For a 4 000 Hz high note,

$$d_{AA} = \frac{\lambda}{2} = \frac{v}{2f} = \frac{343 \text{ m/s}}{8\ 000 \text{ Hz}} = 0.042\ 9 \text{ m} = \boxed{42.9 \text{ mm}}$$

P18.41 (a) The wavelength is $\lambda = \dfrac{v}{f} = \dfrac{343 \text{ m/s}}{261.6/\text{s}} = 1.31 \text{ m}$,



ANS. FIG. P18.41

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so the length of the open pipe vibrating in its simplest (ANA) mode is

$$d_{\text{A to A}} = \frac{1}{2}\lambda = \boxed{0.656 \text{ m}}$$

(b) A closed pipe has (NA) for its simplest resonance, (NANA) for the second, and (NANANA) for the third, equal to 5/4 wavelengths.

Here, the pipe length is $5d_{\text{N to A}} = \frac{5\lambda}{4} = \frac{5}{4}(1.31 \text{ m}) = \boxed{1.64 \text{ m}}$

P18.42 At $T_C = 0.00°\text{C}$, the speed of sound is

$$v = 331 \text{ m/s}\sqrt{1 + \frac{T_C}{273}} = 331 \text{ m/s}$$

(a) For a pipe closed at one end,

$$f_1 = \frac{v}{4L} = \frac{331 \text{ m/s}}{4(4.88 \text{ m})} = \boxed{17.0 \text{ Hz}}$$

(b) For a pipe open at each end,

$$f_1 = \frac{v}{2L} = \frac{331 \text{ m/s}}{2(4.88 \text{ m})} = \boxed{33.9 \text{ Hz}}$$

(c) At $T_C = 20.0°\text{C}$, the speed of sound is $v = 343 \text{ m/s}$.

closed at one end:

$$f_1 = \frac{v}{4L} = \frac{343 \text{ m/s}}{4(4.88 \text{ m})} = \boxed{17.6 \text{ Hz}}$$

open at each end:

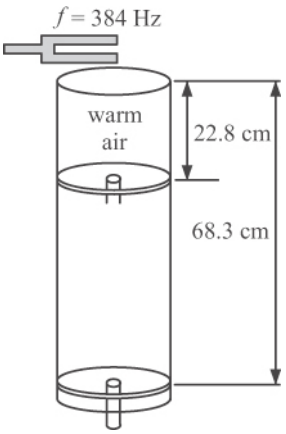
$$f_1 = \frac{v}{2L} = \frac{343 \text{ m/s}}{2(4.88 \text{ m})} = \boxed{35.1 \text{ Hz}}$$

P18.43 For resonance in a narrow tube open at one end,

$$f = n\frac{v}{4L}(n = 1,\ 3,\ 5,\ \ldots)$$

(a) The node–node distance is

$$d_{\text{NN}} = 68.3 \text{ cm} - 22.8 \text{ cm} = 45.5 \text{ cm}$$



ANS. FIG. P18.43

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This distance is equal to half the wavelength, so,

$$v = \lambda f = 2d_{NN} f$$
$$= 2(0.455\text{ m})(384\text{ Hz})$$
$$= \boxed{349\text{ m/s}}$$

(b) Resonance will be established when the tube length has increased by another half wavelength: 68.3 cm + 45.5 = 113.8 = $\boxed{1.14\text{ m}}$

P18.44 The tube acts as a pipe open at one end and closed at the other. Resonance frequencies are odd harmonics. The length corresponding to the fundamental satisfies

$$f_1 = \frac{v}{4L}: \quad L_1 = \frac{v}{4f_1} = \frac{343\text{ m/s}}{4\left(512\text{ s}^{-1}\right)} = 0.167\text{ m}$$

Since $L > 20.0$ cm, the *next* two modes will be observed,

corresponding to $f_2 = \frac{3v}{4L_2}$ and $f_3 = \frac{5v}{4L_3}$,

or $L_2 = \frac{3v}{4f_2} = 3f_1 = \boxed{0.502\text{ m}}$ and $L_3 = \frac{5v}{4f_3} = 5f_1 = \boxed{0.837\text{ m}}$.

P18.45 (a) For the fundamental mode of an open tube,

$$L = \frac{\lambda_1}{2} = \frac{v}{2f_1} = \frac{343\text{ m/s}}{2\left(880\text{ s}^{-1}\right)} = \boxed{0.195\text{ m}}$$

(b) $$v = 331\text{ m/s}\sqrt{1 + \frac{(-5.00)}{273}} = 328\text{ m/s}$$

We ignore the thermal expansion of the metal.

$$f_1 = \frac{v}{\lambda_1} = \frac{v}{2L} = \frac{328\text{ m/s}}{2(0.195\text{ m})} = \boxed{841\text{ Hz}}$$

The flute is flat by a semitone.

P18.46 For a closed box, the resonant frequencies will have nodes at both sides, so the permitted wavelengths will be $\lambda = \frac{2L}{n}$, (n = 1, 2, 3, …),

i.e., $L = \frac{n\lambda}{2} = \frac{nv}{2f}$ and $f = \frac{nv}{2L}$.

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Therefore, with $L = 0.860$ m, $L' = 2.10$ m, and $v = 355$ m/s, the resonant frequencies are

$$f_n = \boxed{n(206 \text{ Hz})} \quad \text{for } L = 0.860 \text{ m for each } n \text{ from 1 to 9}$$

and $$f_n' = \boxed{n(84.5 \text{ Hz})} \quad \text{for } L' = 2.10 \text{ m for each } n \text{ from 2 to 23.}$$

***P18.47** $\frac{\lambda}{2} = d_{AA} = \frac{L}{n}$ or $L = \frac{n\lambda}{2}$ for $n = 1, 2, 3, \ldots$

Since $\lambda = \frac{v}{f}$, $L = n\left(\frac{v}{2f}\right)$ for $n = 1, 2, 3, \ldots$

With $v = 343$ m/s and $f = 680$ Hz,

$$L = n\left(\frac{343 \text{ m/s}}{2(680 \text{ Hz})}\right) = n(0.252 \text{ m}) \quad \text{for } n = 1, 2, 3, \ldots$$

Possible lengths for resonance are:

$$L = \boxed{0.252 \text{ m}, 0.504 \text{ m}, 0.757 \text{ m}, \ldots, n(0.252) \text{ m}}$$

P18.48 (a) The open ends of the tunnel are antinodes, so $d_{AA} = 2\,000 \text{ m}/n$, with $n = 1, 2, 3, \ldots$

Then

$$\lambda = 2d_{AA} = 4\,000 \text{ m}/n$$

and

$$f = \frac{v}{\lambda} = \frac{343 \text{ m/s}}{4\,000 \text{ m}/n} = \boxed{0.085\,8n \text{ Hz, with } n = 1, 2, 3, \ldots}$$

(b) It is a good rule. Any car horn would produce several or many of the closely-spaced resonance frequencies of the air in the tunnel, so it would be greatly amplified. Other drivers might be frightened directly into dangerous behavior, or might blow their horns also.

P18.49 The wavelength of the sound from the tuning fork is $\lambda = \frac{v}{f}$. The cylinder is a pipe open at the top and closed at the water surface; its resonance patterns are AN, ANAN, ANANAN, etc. Resonance occurs each time the height of the air column changes by half a wavelength: $\Delta h = \frac{v}{2f}$. The volume of the pipe between these two water levels is $\pi r^2 \Delta h$, which is also equal to the amount of water that has entered the

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pipe at rate R in a time interval Δt and has filled this volume. Therefore,

$$R\Delta t = \pi r^2 \Delta h = \frac{\pi r^2 v}{2f} \quad \rightarrow \quad \Delta t = \frac{\pi r^2 v}{2Rf}$$

$$\Delta t = \frac{\pi r^2 v}{2Rf} = \frac{\pi (0.050\ 0\ \text{m})^2 (343\ \text{m/s})}{2(1.00\ \text{L/min})(512\ \text{Hz})}\left(\frac{1\ \text{L}}{10^3\ \text{cm}^3}\right)\left(\frac{100\ \text{cm}}{\text{m}}\right)^3$$

$$= (2.63\ \text{min})\left(\frac{60\ \text{s}}{1\ \text{min}}\right) = \boxed{158\ \text{s}}$$

P18.50 The wavelength of the sound from the tuning fork is $\lambda = \frac{v}{f}$. The cylinder is a pipe open at the top and closed at the water surface; its resonance patterns are AN, ANAN, ANANAN, etc. Resonance occurs each time the height of the air column changes by half a wavelength: $\Delta h = \frac{v}{2f}$. The volume of the pipe between these two water levels is $\pi r^2 \Delta h$, which is also equal to the amount of water that has entered the pipe at rate R in a time interval Δt and has filled this volume. Therefore,

$$R\Delta t = \pi r^2 \Delta h = \frac{\pi r^2 v}{2f} \rightarrow \Delta t = \boxed{\frac{\pi r^2 v}{2Rf}}$$

P18.51 For both open and closed pipes, resonant frequencies are equally spaced as numbers. The set of resonant frequencies then must be 650 Hz, 550 Hz, 450 Hz, 350 Hz, 250 Hz, 150 Hz, 50 Hz. These are odd-integer multipliers of the fundamental frequency of $\boxed{50.0\ \text{Hz}}$. Then the pipe length is $d_{\text{NA}} = \frac{\lambda}{4} = \frac{v}{4f} = \frac{343\ \text{m/s}}{4\left(50.0\ \text{s}^{-1}\right)} = \boxed{1.72\ \text{m}}$.

P18.52 For an air column of length 0.730 m, the column may be open ended or closed at one end. For a column open at both ends:

$$f_n = n\frac{v}{2L} \quad \text{where } n = 1, 2, 3,\ldots$$

$$f_n = n\frac{v}{2L} = n\frac{343\ \text{m/s}}{2(0.730\ \text{m})} = n(235\ \text{Hz}) \quad \text{where } n = 1, 2, 3,\ldots$$

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And thus 235 Hz belongs to the harmonic series of an open column (with $n = 1$), but 587 Hz does not match this harmonic series. Similarly, for a column open only at one end:

$$f_n = n\frac{v}{4L}, \quad \text{where } n = 1, 3, 5,\ldots \text{ (only odd harmonics)}$$

$$f_n = n\frac{v}{4L} = n\frac{343 \text{ m/s}}{4(0.730 \text{ m})} = n(117.5 \text{ Hz}), \quad \text{where } n = 1, 3, 5,...$$

and 587 Hz belongs to the harmonic series of a column open at only one end (for $n = 5$), but 235 Hz does not match this harmonic series. Therefore, it is impossible because a single column could not produce both frequencies.

P18.53 (a) The well acts like a pipe open at one end and closed at the other. The normal modes of vibrations of such a pipe are odd harmonics of a fundamental. Call L the depth of the well and v the speed of sound.

Then for some integer n,

$$L = (2n-1)\frac{\lambda_1}{4} = (2n-1)\frac{v}{4f_1} = \frac{(2n-1)(343 \text{ m/s})}{4(51.87 \text{ s}^{-1})}$$

and for the next resonance

$$L = [2(n+1)-1]\frac{\lambda_2}{4} = (2n+1)\frac{v}{4f_2} = \frac{(2n+1)(343 \text{ m/s})}{4(59.85 \text{ s}^{-1})}$$

Thus,

$$\frac{(2n-1)(343 \text{ m/s})}{4(51.87 \text{ s}^{-1})} = \frac{(2n+1)(343 \text{ m/s})}{4(59.85 \text{ s}^{-1})}$$

and we require an *integer* solution to $\frac{2n+1}{59.85} = \frac{2n-1}{51.87}$.

The equation gives $n = 7$, which gives us

$$L = \frac{[2(7)-1](343 \text{ m/s})}{4(51.87 \text{ s}^{-1})} = \frac{[2(7)+1](343 \text{ m/s})}{4(59.85 \text{ s}^{-1})} = \boxed{21.5 \text{ m}}$$

(b) The first harmonic (fundamental frequency) of the well is $f_1 = v/4L = (343 \text{ m/s})/[4(21.5 \text{ m})] = 3.99$ Hz; its pattern is AN. The 3rd harmonic pattern is ANAN, the 5th is ANANAN, etc. We can see that a pattern with n antinodes is the $(2n - 1)$th harmonic. The frequency 51.87 Hz = 13(3.99 Hz) is the 13th harmonic: $13 = 2(7) - 1$, so the standing wave has 7 antinodes.

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Section 18.6 Standing Waves in Rods and Membranes

P18.54 When the rod is clamped at one-quarter of its length, the vibration pattern reads ANANA and the rod length is $L = 2d_{AA} = \lambda$.

Therefore, $$L = \frac{v}{f} = \frac{5\,100 \text{ m/s}}{4\,400 \text{ Hz}} = \boxed{1.16 \text{ m}}$$

P18.55 (a) The mode pattern is ANA, corresponding to the fundamental mode. The length of the rod is one wavelength:

$$f = \frac{v}{2L} = \frac{5\,100}{(2)(1.60)} = \boxed{1.59 \text{ kHz}}$$

(b) Since it is held in the center, there must be a node in the center as well as antinodes at the ends. The even harmonics have an antinode at the center so only $\boxed{\text{the odd harmonics}}$ are present.

(c) The wavelength is the same as in (a):

$$f = \frac{v'}{2L} = \frac{3\,560}{(2)(1.60)} = \boxed{1.11 \text{ kHz}}$$

Section 18.7 Beats: Interference in Time

P18.56 (a) The string could be tuned to either $\boxed{521 \text{ Hz or } 525 \text{ Hz}}$ from this evidence.

(b) Tightening the string raises the wave speed and frequency. If the frequency were originally 521 Hz, the beats would slow down.

Instead, the frequency must have started at 525 Hz to become $\boxed{526 \text{ Hz}}$.

(c) From $f = \frac{v}{\lambda} = \frac{\sqrt{T/\mu}}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$,

$$\frac{f_2}{f_1} = \sqrt{\frac{T_2}{T_1}} \quad \text{and} \quad T_2 = \left(\frac{f_2}{f_1}\right)^2 T_1.$$

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The fractional change that should be made in the tension is then

$$\text{fractional change} = \frac{T_2 - T_1}{T_1} = \frac{T_2}{T_1} - 1 = T_2 = \left(\frac{f_2}{f_1}\right)^2 - 1$$

$$= \left(\frac{523}{526}\right)^2 - 1 = -0.011\,4 = -1.14\%$$

The tension should be $\boxed{\text{reduced by } 1.14\%}$.

P18.57 Combining the velocity and the tension equations $v = f\lambda$ and $v = \sqrt{T/\mu}$, we find that the frequency is

$$f = \sqrt{\frac{T}{\mu\lambda^2}}$$

Since μ and λ are constant, we can apply that equation to both frequencies, and then divide the two equations to get the proportion

$$\frac{f_1}{f_2} = \sqrt{\frac{T_1}{T_2}}$$

With $f_1 = 110$ Hz, $T_1 = 600$ N, and $T_2 = 540$ N we have

$$f_2 = (110 \text{ Hz})\sqrt{\frac{540 \text{ N}}{600 \text{ N}}} = 104.4 \text{ Hz}$$

The beat frequency is

$$f_b = |f_1 - f_2| = 110 \text{ Hz} - 104.4 \text{ Hz} = \boxed{5.64 \text{ beats/s}}$$

P18.58 We use $f' = \left(\frac{v + v_O}{v - v_S}\right) f$. The observer is stationary, so $v_0 = 0$.

For the approaching train $v_s = +8.00$ m/s; the frequency arriving at the observer is

$$f_1' = \left(\frac{v}{v - v_S}\right) f = \left(\frac{343 \text{ m/s}}{343 \text{ m/s} - 8.00 \text{ m/s}}\right) f = \left(\frac{343 \text{ m/s}}{335 \text{ m/s}}\right) f$$

For the receding train is $v_s = -8.00$ m/s; the frequency arriving at the observer is

$$f_2' = \left(\frac{v}{v - v_S}\right) f = \left(\frac{343 \text{ m/s}}{343 \text{ m/s} - (-8.00 \text{ m/s})}\right) f = \left(\frac{343 \text{ m/s}}{351 \text{ m/s}}\right) f$$

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The beat frequency between the waves emanating from the trains is

$$f_{\text{beat}} = |f_1 - f_2|$$

and, because the receding train produces a lower frequency,

$$f_1' - f_2' = f_{\text{beat}} \quad \rightarrow \quad \left(\frac{343 \text{ m/s}}{335 \text{ m/s}}\right) f - \left(\frac{343 \text{ m/s}}{351 \text{ m/s}}\right) f = 4.00 \text{ Hz}$$

$$\left[\left(\frac{343 \text{ m/s}}{335 \text{ m/s}}\right) - \left(\frac{343 \text{ m/s}}{351 \text{ m/s}}\right)\right] f = 4.00 \text{ Hz} \quad \rightarrow \quad f = \boxed{85.7 \text{ Hz}}$$

P18.59 The source moves toward the wall:

$$v_s = +v_{\text{student}}, \; v_0 = 0, \quad \text{and} \quad f' = f\frac{(v + v_o)}{(v - v_s)} = f\frac{v}{(v - v_{\text{student}})}.$$

The wall acts as stationary source, reflecting the wave of frequency f'. The observe moves toward the source: $v_s = 0$, $v_0 = +v_{\text{student}}$, and

$$f'' = f'\frac{(v + v_o)}{(v - v_s)} = f'\frac{(v + v_s)}{v} = f\frac{v}{(v - v_{\text{student}})}\frac{(v + v_{\text{student}})}{v}$$

$$= f\frac{(v + v_{\text{student}})}{(v - v_{\text{student}})}$$

(a) When the student walks toward the wall f'' is larger than f; the beat frequency is

$$f_b = |f'' - f| = f\frac{(v + v_{\text{student}})}{(v - v_{\text{student}})} - f = f\left[\frac{(v + v_{\text{student}})}{(v - v_{\text{student}})} - 1\right]$$

$$= f\frac{2v_{\text{student}}}{(v - v_{\text{student}})}$$

$$f_b = (256 \text{ Hz})\frac{2(1.33 \text{ m/s})}{(343 \text{ m/s} - 1.33 \text{ m/s})} = \boxed{1.99 \text{ Hz}}$$

(b) When he is moving away from the wall, the sign of v_{student} changes and f'' is smaller than f:

$$f_b = |f'' - f| = f - f\frac{(v - v_{\text{student}})}{(v + v_{\text{student}})} = f\left[1 - \frac{(v - v_{\text{student}})}{(v + v_{\text{student}})}\right]$$

$$= f\frac{2v_{\text{student}}}{(v + v_{\text{student}})}$$

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Solving for v_{student} gives

$$v_{\text{student}} = \frac{f_b v}{2f - f_b} = \frac{(5\text{ Hz})(343\text{ m/s})}{(2)(256\text{ Hz}) - 5\text{ Hz}} = \boxed{3.38\text{ m/s}}$$

Section 18.8 Nonsinusoidal Wave Patterns

***P18.60** We list the frequencies of the harmonics of each note in Hz:

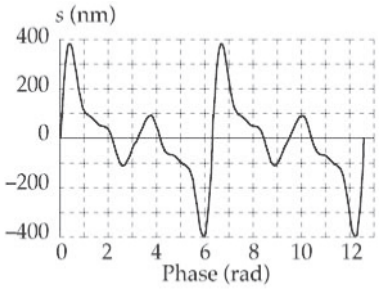
	Harmonic				
Note	1	2	3	4	5
A	440.00	880.00	1 320.0	1 760.0	2 200.0
C#	554.37	1 108.7	1 663.1	2 217.5	2 771.9
E	659.26	1 318.5	1 977.8	2 637.0	3 296.3

The second harmonic of E is close the the third harmonic of A, and the fourth harmonic of C# is close to the fifth harmonic of A.

P18.61 We evaluate

$$s = 100\sin\theta + 157\sin 2\theta + 62.9\sin 3\theta + 105\sin 4\theta + 51.9\sin 5\theta + 29.5\sin 6\theta + 25.3\sin 7\theta$$

where s represents particle displacement in nanometers and θ represents the phase of the wave in radians. As θ advances by 2π, time advances by (1/523) s. The resultant waveform is shown below in ANS. FIG. P18.61.



ANS. FIG. P18.61

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Additional Problems

***P18.62** (a) The fundamental wavelength of the pipe open at both ends is $\lambda = 2L = v / f_1$. Since the speed of sound is 331 m/s at 0°C, the length of the pipe is

$$L = \frac{v}{2f_1} = \frac{331 \text{ m/s}}{2(300 \text{ Hz})} = \boxed{0.552 \text{ m}}$$

(b) At T = 30.0°C = 303 K,

$$v = (331 \text{ m/s})\sqrt{\frac{T_K}{273}} = (331 \text{ m/s})\sqrt{\frac{303}{273}} = 349 \text{ m/s}$$

and

$$f_1 = \frac{v}{\lambda_1} = \frac{v}{2L} = \frac{349 \text{ m/s}}{2(0.552 \text{ m})} = \boxed{316 \text{ Hz}}$$

***P18.63** The second standing wave mode of the air in the pipe reads ANAN,

with $d_{\text{NA}} = \frac{\lambda}{4} = \frac{1.75 \text{ m}}{3}$,

so $\lambda = 2.33 \text{ m}$

and $f = \frac{v}{\lambda} = \frac{343 \text{ m/s}}{2.33 \text{ m}} = 147 \text{ Hz}.$

For the string, λ and v are different but f is the same.

$$\frac{\lambda}{2} = d_{\text{NN}} = \frac{0.400 \text{ m}}{2}$$

so $\lambda = 0.400 \text{ m}.$

$$v = \lambda f = (0.400 \text{ m})(147 \text{ Hz}) = 58.8 \text{ m/s} = \sqrt{\frac{T}{\mu}}$$

$$T = \mu v^2 = \left(9.00 \times 10^{-3} \text{ kg/m}\right)(58.8 \text{ m/s})^2 = \boxed{31.1 \text{ N}}$$

P18.64 The beat frequency between the waves emanating from the two strings is

$$f_{beat} = | f_1 - f_2 |$$

and, because the decrease in tension causes the second frequency to be lower,

$$f_2 = f_1 - f_{beat} = (150 \text{ Hz}) - (4 \text{ Hz}) = \boxed{146 \text{ Hz}}$$

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P18.65 At point D, the distance of the ship from point A is

$$d_1 = \sqrt{d_2^2 + (800\text{ m})^2} = \sqrt{(600\text{ m})^2 + (800\text{ m})^2} = 1\,000\text{ m}$$

Since destructive interference occurs for the first time when the ship reaches D, it is necessary that $d_1 - d_2 = \lambda/2$ or

$$\lambda = 2(d_1 - d_2) = 2(1\,000\text{ m} - 600\text{ m}) = \boxed{800\text{ m}}$$

P18.66 According to Equation 18.6, the natural frequencies of vibration of a string fixed at both ends are given by

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}} = \frac{n}{2(2.00\text{ m})}\sqrt{\frac{20.0\text{ N}}{\left(\frac{0.100\text{ kg}}{2.00\text{ m}}\right)}} = n(5.00\text{ Hz})$$

where $n = 1, 2, 3, \ldots$

(a) $f_1 = \boxed{5.0\text{ Hz}}$, $f_2 = \boxed{10.0\text{ Hz}}$, $f_3 = \boxed{15.0\text{ Hz}}$

(b) This could be any mode that has a node 0.400 m from an end. If $D = 0.400$ m is the distance between adjacent nodes (the distance across a pair of nodes), $d_{NN} = D = \lambda/2$, and its wavelength is 0.800 m:

$$\frac{\lambda}{2} = D \quad \rightarrow \quad \lambda = 2D = 2(0.400\text{ m})$$

$$\lambda = \frac{2L}{n} \quad \rightarrow \quad n = \frac{2L}{\lambda} = \frac{2L}{2D} = \frac{L}{D} = \frac{2.00\text{ m}}{0.400\text{ m}} = 5$$

This mode corresponds to the 5th harmonic: $f_5 = 5(5.00\text{ Hz}) = 25.0\text{ Hz}$. But D could be the distance across two pairs of nodes (from node to node to node), $d_{NNN} = D = 2(\lambda/2)$, or three pairs, d_{NNN}, or across N pairs of nodes:

$$N\frac{\lambda}{2} = D \quad \rightarrow \quad \lambda = 2D/N$$

then,

$$n = \frac{2L}{\lambda} = \frac{2L}{(2D/N)} = N\frac{L}{D} = N\frac{2.00\text{ m}}{0.400\text{ m}} = 5N$$

and so on, corresponding to the 10th, or the 15th harmonic, etc.

The frequency could be the fifth state at 25.0 Hz or any integer multiple, such as the tenth state at 50.0 Hz, the fifteenth state at 75.0 Hz, and so on.

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P18.67 When the string is plucked, nodes occur on the ends because they are fixed. The plucked guitar string vibrates in its fundamental mode with a wavelength equal to twice the length of the string. For the 2 349-Hz note, the length of the vibrating string is $L = 21.4$ cm. For the 2 217-Hz note, the length of the vibrating string is $L + x$, where x is the distance to the next fret. We wish to solve for x.

We assume the wave speed is the same on each string. Compare the frequencies to the lengths of vibrating string:

$$f_1 = 2349 \text{ Hz} = \frac{v}{2L}$$

$$f_2 = 2217 \text{ Hz} = \frac{v}{2(L+x)}$$

Taking the ratio,

$$\frac{f_1}{f_2} = \frac{L+x}{L} = 1 + \frac{x}{L}$$

$$x = L\left(\frac{f_1}{f_2} - 1\right) = (21.4 \text{ cm})\left(\frac{2\ 349 \text{ Hz}}{2\ 217 \text{ Hz}} - 1\right) = \boxed{1.27 \text{ cm}}$$

P18.68 (a) The frequency of the normal mode produces a sound wave of the same frequency. For the same frequency, wavelength is proportional to wave speed. On the string, the wave speed is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{(48.0 \text{ N})}{\left(\frac{4.80 \times 10^{-3} \text{ kg}}{2.00 \text{ m}}\right)}} = 141 \text{ m/s}$$

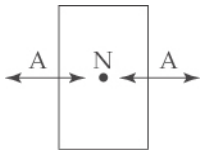
which is smaller than the speed of sound (343 m/s).

The wavelength in air of the sound produced by the string is $\boxed{\text{larger}}$ because the wave speed is larger.

(b) $$\frac{\lambda_{\text{air}}}{\lambda_{\text{string}}} = \frac{v_{\text{air}}/f}{v_{\text{string}}/f} = \frac{v_{\text{air}}}{v_{\text{string}}} = \frac{343 \text{ m/s}}{141 \text{ m/s}} = \boxed{2.43}$$

P18.69 $d_{AA} = \dfrac{\lambda}{2} = 7.05 \times 10^{-3}$ m is the distance between antinodes. Then $\lambda = 14.1 \times 10^{-3}$ m,

and $f = \dfrac{v}{\lambda} = \dfrac{3.70 \times 10^3 \text{ m/s}}{14.1 \times 10^{-3} \text{ m}} = \boxed{2.62 \times 10^5 \text{ Hz}}$,



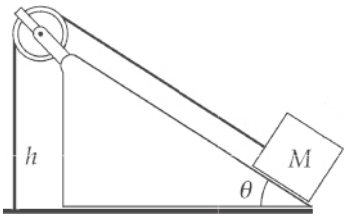
ANS. FIG. P18.69

The crystal can be tuned to vibrate at 2^{18} Hz, so that binary counters can derive from it a signal at precisely 1 Hz.

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P18.70 (a) The particle under constant acceleration model

(b) Waves under boundary conditions model



ANS. FIG. P18.70

(c) For the block:

$$\sum F_x = T - Mg\sin\theta = 0$$

so $T = \boxed{Mg\sin\theta}$.

(d) The length of the section of string parallel to the incline is $\dfrac{h}{\sin\theta}$.

The total length of the string is then

$$L = \frac{h}{\sin\theta} + h = \frac{h}{\sin\theta} + \frac{h\sin\theta}{\sin\theta} = \boxed{h\left(\frac{1+\sin\theta}{\sin\theta}\right)}$$

(e) The mass per unit length of the string is

$$\mu = \frac{m}{L} = \frac{m}{h\left(\dfrac{1+\sin\theta}{\sin\theta}\right)} = \boxed{\frac{m\sin\theta}{h(1+\sin\theta)}}$$

(f) The speed of waves in the string is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg\sin\theta}{\left[\dfrac{m\sin\theta}{h(1+\sin\theta)}\right]}} = \boxed{\sqrt{\frac{Mgh}{m}(1+\sin\theta)}}$$

(g) The fundamental mode vibrates at the lowest frequency. In the fundamental mode, the segment of length h vibrates as one loop. The distance between adjacent nodes is then $d_{NN} = \dfrac{\lambda}{2} = h,$ so the wavelength is $\lambda = 2h.$

The frequency is $f = \dfrac{v}{\lambda} = \dfrac{1}{2h}\sqrt{\dfrac{Mgh}{m}(1+\sin\theta)} = \boxed{\sqrt{\dfrac{Mg}{4mh}(1+\sin\theta)}}.$

(h) $$f = \sqrt{\frac{Mg}{4mh}(1+\sin\theta)} = \sqrt{\frac{(1.50\text{ kg})(9.80\text{ m/s}^2)}{4(0.750\times10^{-3}\text{ kg})(0.500\text{ m})}(1+\sin 30.0^\circ)}$$

$$= \boxed{121\text{ Hz}}$$

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(i) The fundamental mode has a wavelength twice the length of the sloped section of string, $\lambda = 2\dfrac{h}{\sin\theta}$. Its frequency is

$$f = \frac{v}{\lambda} = \frac{1}{\left(2\dfrac{h}{\sin\theta}\right)}\sqrt{\frac{Mgh}{m}\left(1+\sin\theta\right)} = \sin\theta\sqrt{\frac{Mg}{4mh}\left(1+\sin\theta\right)}$$

$$f = \sin 30.0°\,(121\text{ Hz}) = \boxed{60.6\text{ Hz}}$$

***P18.71** For the wire, $\mu = \dfrac{0.010\ 0\text{ kg}}{2.00\text{ m}} = 5.00\times10^{-3}\text{ kg/m}$:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{200\text{ kg}\cdot\text{m/s}^2}{5.00\times10^{-3}\text{ kg/m}}} = 200\text{ m/s}$$

If it vibrates in its simplest state, $d_{NN} = 2.00\text{ m} = \dfrac{\lambda}{2}$:

$$f = \frac{v}{\lambda} = \frac{200\text{ m/s}}{4.00\text{ m}} = 50.0\text{ Hz}$$

(a) The tuning fork can have frequencies $\boxed{45.0\text{ Hz or }55.0\text{ Hz.}}$

(b) If $f = 45.0$ Hz, and $v = f\lambda = \left(45.0\text{ s}^{-1}\right)(4.00\text{ m}) = 180\text{ m/s},$ then

$$T = v^2\mu = (180\text{ m/s})^2\left(5.00\times10^{-3}\text{ kg/m}\right) = \boxed{162\text{ N}}$$

or if $f = 55.0$ Hz,

$$T = v^2\mu = f^2\lambda^2\mu = \left(55.0\text{ s}^{-1}\right)^2(4.00\text{ m})^2\left(5.00\times10^{-3}\text{ kg/m}\right)$$

$$= \boxed{242\text{ N}}$$

***P18.72** We are told that the man's ears are at the same level as the lower speaker. Sound from the upper speaker is delayed by traveling the extra distance $\Delta r = \sqrt{L^2 + d^2} - L.$

He hears a minimum when $\Delta r = (2n-1)\left(\dfrac{\lambda}{2}\right),$ with $n = 1, 2, 3, \ldots$

Then,

$$\sqrt{L^2 + d^2} - L = \left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right)$$

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$$\sqrt{L^2 + d^2} = \left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right) + L$$

$$L^2 + d^2 = \left(n - \frac{1}{2}\right)^2\left(\frac{v}{f}\right)^2 + 2\left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right)L + L^2$$

$$d^2 - \left(n - \frac{1}{2}\right)^2\left(\frac{v}{f}\right)^2 = 2\left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right)L \qquad \textbf{[1]}$$

Equation [1] gives the distances from the lower speaker at which the man will hear a minimum. The path difference Δr starts from nearly zero when the man is very far away and increases to d when $L = 0$.

(a) The number of minima he hears is the greatest integer value for which $L \geq 0$. This is the same as the greatest integer solution to

$$d \geq \left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right)$$

or

$$\boxed{\begin{aligned}\text{number of minima heard } &= n_{\max} \\ &= \text{greatest integer } \leq d\left(\frac{f}{v}\right) + \frac{1}{2}\end{aligned}}$$

(b) From equation [1], the distances at which minima occur are given by

$$\boxed{L_n = \frac{d^2 - \left(n - \frac{1}{2}\right)^2\left(\frac{v}{f}\right)^2}{2\left(n - \frac{1}{2}\right)\left(\frac{v}{f}\right)}, \text{ where } n = 1,\ 2,\ \ldots,\ n_{\max}}$$

P18.73 (a) The tension on the string defines the wave velocity on the string, and thus also the frequencies, wavelengths, and number of nodes of the standing waves. The tension on the string in Figure 18.11a is:

$T_1 = mg$, where m is the mass of the sphere.

The tension on the string in Figure 18.11b, must also include the buoyant force on the sphere:

$$T_2 = mg - B = mg - \rho_{\text{water}} g V_{\text{sphere}} = mg - \rho_{\text{water}} g\left(\frac{4}{3}\pi r^3\right)$$

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Notice that the number of antinodes n is exactly the number of half wavelengths of standing waves on the string (i.e. there are two antinodes (and one full wavelength) on the string in Figure 18.11a, and there are five antinodes (and two and a half full wavelengths) in Figure 18.11b). From Equations 18.5 and 18.6 we have

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2L} = \frac{n}{2L}\sqrt{\frac{T}{\mu}},\ n = 1,\ 2,\ 3,\ 4,...$$

The frequency of oscillation is the same in both cases because it is defined by the moving blade to the left. In addition, neither the total length of the string L nor the string density μ changes between the two cases:

$$f = \frac{n_1}{2L}\sqrt{\frac{T_1}{\mu}} \quad \text{and} \quad f = \frac{n_2}{2L}\sqrt{\frac{T_2}{\mu}}$$

therefore,

$$2Lf\sqrt{\mu} = n_1\sqrt{T_1} = n_2\sqrt{T_2}$$

Or equivalently,

$$T_2 = \left(\frac{n_1}{n_2}\right)^2 T_1 = \left(\frac{n_1}{n_2}\right)^2 mg$$

But we have already obtained the value for tension above in terms of the buoyant force and thus the radius of the sphere.

$$T_2 = \left(\frac{n_1}{n_2}\right)^2 T_1 = \left(\frac{n_1}{n_2}\right)^2 mg = mg - \rho_{\text{water}} g\left(\frac{4}{3}\pi r^3\right)$$

The radius of the sphere r may now be solved in terms of the number of antinodes n_2 (and the other parameters, n_1, m, g, and ρ_{water} which are all constants, or already defined in the problem).

$$\rho_{\text{water}} g\left(\frac{4}{3}\pi r^3\right) = mg\left(1 - \frac{n_1^2}{n_2^2}\right) \quad \rightarrow \quad r^3 = \frac{3m}{4\pi\rho_{\text{water}}}\left(1 - \frac{n_1^2}{n_2^2}\right)$$

solving for r gives

$$r = \left[\left(\frac{3m}{4\pi\rho_{\text{water}}}\right)\left(1 - \frac{n_1^2}{n_2^2}\right)\right]^{1/3}$$

$$= \left\{\left[\frac{3(2.00\ \text{kg})}{4\pi\left(10^3\ \text{kg/m}^3\right)}\right]\left(1 - \frac{4}{n^2}\right)\right\}^{1/3} = 0.078\ 2\left(1 - \frac{4}{n^2}\right)^{1/3}$$

where r is in meters.

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(b) Because the factor inside the cube root

$$\left(1-\frac{4}{n^2}\right)^{1/3}$$

will be either zero or negative, which are each meaningless results, for $n = 1$ and 2, the minimum allowed value of n for a sphere of nonzero size is $n = \boxed{3}$.

(c) Because the mass of the sphere is held constant, while its radius (and thus also volume and density) is changed, there will reach a point where the density of the sphere reaches the density of water, and thus the sphere will float, so that it will no longer be fully immersed in the water. After this point, the sphere will float on the water, and will not produce further standing waves.

The limiting condition is $\rho_{\text{sphere}} = \rho_{\text{water}} = 1.00\times10^3 \text{ kg/m}^3$.

But $\rho_{\text{sphere}} = \dfrac{m}{V} = \dfrac{m}{\frac{4}{3}\pi r^3}$ which may be rearranged to solve for r.

$$\frac{4}{3}\pi r^3 = \frac{m}{\rho_{\text{sphere}}} = \frac{m}{\rho_{\text{water}}} \quad \rightarrow \quad r = \left(\frac{3m}{4\pi\rho_{\text{water}}}\right)^{1/3}$$

and substituting in numerically:

$$r = \left(\frac{3m}{4\pi\rho_{\text{water}}}\right)^{1/3} = \left(\frac{3(2.00 \text{ kg})}{4\pi\left(1.0\times10^3 \text{ kg/m}^3\right)}\right)^{1/3}$$

$$= \left(4.766\times10^{-4} \text{ m}^3\right)^{1/3} = \boxed{0.078\ 2 \text{ m}}$$

is the limiting (maximum) radius for which the sphere will stay totally immersed.

(d) The sphere floats on the water.

P18.74 (a) The wavelength is twice the length of string from the top end to the yo-yo: $\lambda = 2L$. The length L changes in time because the yo-yo is a particle under constant acceleration: $L = L_0 + \frac{1}{2}at^2$, where L_0 is the length of the string at $t = 0$ and a is the acceleration of the yo-yo. Therefore,

$$\boxed{\frac{d\lambda}{dt} = \frac{d}{dt}(2L) = \frac{d}{dt}\left[2\left(L_0 + \frac{1}{2}at^2\right)\right] = 2at}$$

$$\boxed{= 2\left(0.800 \text{ m/s}^2\right)(1.20 \text{ s}) = 1.92 \text{ m/s}}$$

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(b) For the second harmonic, the wavelength is equal to the length of the string. Therefore,

$$\frac{d\lambda}{dt} = \frac{d}{dt}L = \frac{d}{dt}\left(L_0 + \frac{1}{2}at^2\right) = at$$

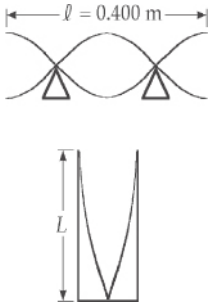
$$= \left(0.800\ \text{m/s}^2\right)(1.20\ \text{s})$$

$$= \boxed{0.960\ \text{m/s, half as much as for the first harmonic}}.$$

(c) Yes. A yo-yo of different mass will hold the string under different tension to make each string wave vibrate with a different frequency, but the geometrical argument given in part (a) still applies to the wavelength.

(d) Yes, for the same reason as in (c): the geometrical argument given in part (b) still applies to the wavelength.

P18.75 $f = 87.0$ Hz. The speed of sound in air is $v_a = 343$ m/s.



ANS. FIG. P18.75

(a) The pattern on the bar (see upper figure at right) is ANANA, corresponding to the second harmonic. The wavelength on the bar is $\lambda_b = L$,

$$v = f\lambda_b = \left(87.0\ \text{s}^{-1}\right)(0.400\ \text{m})$$

$$= \boxed{34.8\ \text{m/s}}$$

(b) With $\lambda_a = 4L$ and $v_a = \lambda_a f$,

$$L = \frac{v_a}{4f} = \frac{343\ \text{m/s}}{4\left(87.0\ \text{s}^{-1}\right)} = \boxed{0.986\ \text{m}}$$

P18.76 (a) $$\mu = \frac{5.50\times10^{-3}\ \text{kg}}{0.860\ \text{m}} = 6.40\times10^{-3}\ \text{kg/m}$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1.30\ \text{kg}\cdot\text{m/s}^2}{6.40\times10^{-3}\ \text{kg/m}}} = \boxed{14.3\ \text{m/s}}$$

(b) The distance between a node and its adjacent antinode is one-quarter of a wavelength. In order for there to be a node at the bottom and an antinode at the top, the string can contain only an odd number of node-antinode pairs.

The simplest pattern is (top to bottom) AN = one node-antinode pair:

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$$\frac{\lambda_1}{4} = L = \boxed{86.0 \text{ cm}}$$

The next simplest pattern is ANAN = AN + NA + AN = three node-antinode pairs:

$$3\frac{\lambda_3}{4} = L \quad \rightarrow \quad \frac{\lambda_3}{4} = \frac{L}{3} = \boxed{28.7 \text{ cm}}$$

The next simplest pattern is ANANAN = AN + NA + AN + NA + AN = five node-antinode pairs:

$$5\frac{\lambda_5}{4} = L \quad \rightarrow \quad \frac{\lambda_5}{4} = \frac{L}{5} = \boxed{17.2 \text{ cm}}$$

(c) The distance between node and an antinode is $\frac{\lambda}{4}$. The corresponding frequency is

$$f_n = \frac{v}{4\left(\frac{\lambda_n}{4}\right)}: \qquad \begin{cases} f_1 = \dfrac{v}{4\left(\frac{\lambda_1}{4}\right)} = \dfrac{14.3 \text{ m/s}}{4(0.860 \text{ m})} = \boxed{4.14 \text{ Hz}} \\ f_3 = \dfrac{v}{4\left(\frac{\lambda_3}{4}\right)} = \dfrac{14.3 \text{ m/s}}{4(0.287 \text{ m})} = \boxed{12.4 \text{ Hz}} \\ f_5 = \dfrac{v}{4\left(\frac{\lambda_5}{4}\right)} = \dfrac{14.3 \text{ m/s}}{4(0.172 \text{ m})} = \boxed{20.7 \text{ Hz}} \end{cases}$$

P18.77 We consider velocities of approach and of recession separately in the Doppler equation, after we observe from our beat equation $f_b = |f_1 - f_2| = |f - f'|$ that the moving train must have an apparent frequency of either $f' = 182$ Hz or $f' = 178$ Hz.

We let v_t represent the magnitude of the train's velocity. If the train is moving away from the station, the apparent frequency is 178 Hz, lower, as described by

$$f' = \frac{v}{v + v_t}$$

and the train is **moving away** at

$$v_t = v\left(\frac{f}{f'} - 1\right) = (343 \text{ m/s})\left(\frac{180 \text{ Hz}}{178 \text{ Hz}} - 1\right) = 3.85 \text{ m/s}$$

If the train is pulling into the station, then the apparent frequency is 182 Hz. Again from the Doppler shift,

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$$f' = \frac{v}{v - v_s}$$

The train is **approaching** at

$$v_s = v\left(1 - \frac{f}{f'}\right) = (343 \text{ m/s})\left(1 - \frac{180 \text{ Hz}}{182 \text{ Hz}}\right)$$

$$v_s = 3.77 \text{ m/s}$$

The moving train has a velocity of either 3.85 m/s away from the station or 3.77 m/s toward the station.

***P18.78** (a) Use the Doppler formula:

$$f' = f\frac{(v \pm v_0)}{(v \mp v_s)}$$

with $f_1' =$ frequency of the speaker in front of student and

$f_2' =$ frequency of the speaker behind the student.

$$f_1' = (456 \text{ Hz})\frac{(343 \text{ m/s} + 1.50 \text{ m/s})}{(343 \text{ m/s} - 0)} = 458 \text{ Hz}$$

$$f_2' = (456 \text{ Hz})\frac{(343 \text{ m/s} - 1.50 \text{ m/s})}{(343 \text{ m/s} + 0)} = 454 \text{ Hz}$$

Therefore, $f_b = f_1' - f_2' =$ 3.99 Hz.

(b) The waves broadcast by both speakers have

$$\lambda = \frac{v}{f} = \frac{343 \text{ m/s}}{456 \text{ s}^{-1}} = 0.752 \text{ m}$$

The standing wave between them has $d_{AA} = \frac{\lambda}{2} = 0.376$ m.

The student walks from one maximum to the next in time

$\Delta t = \frac{0.376 \text{ m}}{1.50 \text{ m/s}} = 0.251$ s, so the frequency at which she hears maxima is

$$f = \frac{1}{T} = \frac{1}{0.251 \text{ s}} = \boxed{3.99 \text{ Hz}}$$

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***P18.79** As in Problem 32, we let $m = \rho V$ represent the mass of the copper cylinder. The original tension in the wire is $T_1 = mg = \rho Vg$. The water exerts a buoyant force $\rho_{\text{water}}(nV)g$ on the copper object, where n is the fraction of the object that is submerged, to reduce the tension to

$$T_2 = \rho Vg - \rho_{\text{water}}(nV)g = (\rho - n\rho_{\text{water}})Vg$$

The speed of a wave on the string changes from $\sqrt{\frac{T_1}{\mu}}$ to $\sqrt{\frac{T_2}{\mu}}$. The frequency changes from

$$f_1 = \frac{v_1}{\lambda} = \left(\frac{1}{\lambda}\right)\sqrt{\frac{T_1}{\mu}} \quad \text{to} \quad f_2 = \left(\frac{1}{\lambda}\right)\sqrt{\frac{T_2}{\mu}}$$

where we assume $\lambda = 2L$ is constant.

Then

$$\frac{f_2}{f_1} = \sqrt{\frac{T_2}{T_1}}$$

and

$$f_2 = f_1\sqrt{\frac{\rho - n\rho_{\text{water}}}{\rho}}$$

The frequency decreases as the fraction of the object that is submerged increases, with the lowest frequency occurring when the object is completely submerged, or $n = 1$:

$$f_2 = f_1\sqrt{\frac{\rho - n\rho_{\text{water}}}{\rho}} = (300\text{ Hz})\sqrt{\frac{8.92 - (1.00)1.00}{8.92}}$$

$$= (300\text{ Hz})\sqrt{\frac{7.92}{8.92}} = \boxed{283\text{ Hz}}$$

P18.80 (a) Since the first node is at the weld, the wavelength in the thin wire is $2L$ or 80.0 cm. The frequency and tension are the same in both sections, so

$$f = \frac{1}{2L}\sqrt{\frac{T}{\mu}} = \frac{1}{2(0.400\text{ m})}\sqrt{\frac{4.60\text{ N}}{2.00\times10^{-3}\text{ kg/m}}} = \boxed{59.9\text{ Hz}}$$

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(b) As the thick wire is twice the diameter, the linear density is 4 times that of the thin wire, or $\mu' = 8.00\ \text{g/m}$.

so
$$L' = \frac{1}{2f}\sqrt{\frac{T}{\mu'}}$$
$$= \left[\frac{1}{(2)(59.9\ \text{s}^{-1})}\right]\sqrt{\frac{4.60\ \text{N}}{8.00\times10^{-3}\ \text{kg/m}}} = \boxed{20.0\ \text{cm}}$$

or half the length of the thin wire.

P18.81 The wavelength stays constant at $\lambda_1 = 2L$ while the wavespeed rises according to

$$v = (T/\mu)^{1/2} = [(15.0 + 10.0t/3.50)/\mu]^{1/2}$$

so the frequency rises as $f = v/\lambda = [(15.0 + 10.0t/3.50)/\mu]^{1/2}/2L$.

The number of cycles is $N = dt/T = f\,dt$ in each incremental bit of time, or altogether

$$N = \frac{1}{2L\sqrt{\mu}}\int_0^{3.5}\left(15.0 + \frac{10.0}{3.50}t\right)^{1/2}dt$$
$$= \frac{1}{2L\sqrt{3.50\mu}}\int_0^{3.5}(52.5 + 10.0t)^{1/2}dt$$
$$N = \frac{1}{2L\sqrt{3.50\mu}}\frac{1}{10.0(3/2)}(52.5 + 10.0t)^{3/2}\Big|_0^{3.5}$$
$$= \frac{1}{30L\sqrt{3.50\mu}}(52.5 + 10.0t)^{3/2}\Big|_0^{3.5}$$
$$N = \frac{1}{30.0(0.480\ \text{m})\sqrt{3.50\left(1.60\times10^{-3}\ \text{kg/m}\right)}} \times\left[(52.5 + 35.0)^{3/2} - (52.5)^{3/2}\right]$$
$$N = \boxed{407\ \text{cycles}}$$

P18.82 We use the basic relationship $f = \dfrac{n}{2L}\sqrt{\dfrac{T}{\mu}}$.

(a) Changing the length does not change the tension or the mass per unit length, so the wave speed is the same.

$$\frac{f'}{f} = \frac{L}{L'} = \frac{L}{2L} = \frac{1}{2}$$

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The $\boxed{\text{frequency should be halved}}$ to get the same number of antinodes for twice the length.

(b) $\dfrac{n'}{n} = \sqrt{\dfrac{T}{T'}}$ so $\dfrac{T'}{T} = \left(\dfrac{n}{n'}\right)^2 = \left[\dfrac{n}{n+1}\right]^2$

The tension must be $T' = \boxed{\left[\dfrac{n}{n+1}\right]^2 T}$.

(c) $\dfrac{f'}{f} = \dfrac{n'L}{nL'}\sqrt{\dfrac{T'}{T}}$ so $\dfrac{T'}{T} = \left(\dfrac{nf'L'}{n'fL}\right)^2 = \left[\left(\dfrac{n}{n'}\right)\left(\dfrac{f'}{f}\right)\left(\dfrac{L'}{L}\right)\right]^2$

$\dfrac{T'}{T} = \left(\dfrac{3}{2\cdot 2}\right)^2 \quad \rightarrow \quad \boxed{\dfrac{T'}{T} = \dfrac{9}{16}}$ to get twice as many antinodes.

P18.83 We look for a solution of the form

$$5.00 \sin(2.00x - 10.0t) + 10.0 \cos(2.00x - 10.0t)$$
$$= A \sin(2.00x - 10.0t + \phi)$$
$$= A \sin(2.00x - 10.0t)\cos\phi + A \cos(2.00x - 10.0t)\sin\phi$$

This will be true if both $5.00 = A\cos\phi$ and $10.0 = A\sin\phi$, requiring

$$(5.00)^2 + (10.0)^2 = A^2 \quad \rightarrow \quad A = 11.2, \text{ and}$$

$$\tan\phi = \frac{10.0}{5.00} = 2.00 \quad \rightarrow \quad \phi = 63.4°$$

(a) $\boxed{\text{From above, we were able to find values for } A \text{ and } \phi\text{; therefore, the resultant wave is sinusoidal.}}$

(b) From above $\boxed{A = 11.2}$ and $\boxed{\phi = 63.4°}$.

P18.84 The speed of sound at Celsius temperature T_C is

$$v = (331 \text{ m/s})\sqrt{1 + \frac{T_C}{273°\text{C}}}$$

At 20.0°C, the speed of sound is 343 m/s.

(a) For a pipe open at both ends, the fundamental frequency ($n = 1$) is

$$f_1 = \frac{v}{2L} \quad \rightarrow \quad L = \frac{v}{2f_1} = \frac{343 \text{ m/s}}{2(261.6 \text{ Hz})} = \boxed{0.656 \text{ m}}$$

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(b) The speed of sound in the colder room is smaller because the temperature is lower. The fundamental frequency of the pipe played in that room, call it f'_1, is smaller because the frequency of a standing wave is proportional to the wave speed. The beat frequency is

$$f_{\text{beat}} = \left| f'_1 - f'_1 \right| = f_1 - f'_1$$

which gives

$$f_1 = f_1 - f_{\text{beat}} = 261.6 - 3.00 = 258.6 \text{ Hz}$$

because $f_1 = 261.6$ Hz is larger than f'_1.

The lengths of the flutes are the same, so compare frequencies and wave speeds:

$$f_1 = \frac{v}{2L} \quad \rightarrow \quad \frac{f'_1}{f_1} = \frac{v'}{v}$$

Solving for the wave speed gives

$$v' = v\frac{f'_1}{f_1} = (343 \text{ m/s})\left(\frac{258.6 \text{ Hz}}{261.6 \text{ Hz}}\right) = 339 \text{ m/s}$$

The wave speed depends on the temperature:

$$v = (331 \text{ m/s})\sqrt{1 + \frac{T_C}{273°}} = 339 \text{ m/s}$$

Solving for the temperature gives

$$T_C = 273°\left[\left(\frac{339 \text{ m/s}}{331 \text{ m/s}}\right)^2 - 1\right] = \boxed{13.5°\text{C}}$$

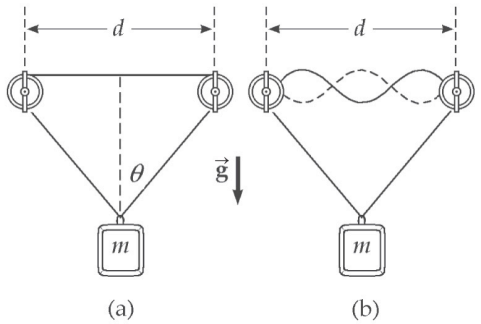
P18.85 (a) Let θ represent the angle each slanted rope makes with the vertical. In the diagram, observe that:

$$\sin\theta = \frac{1.00 \text{ m}}{1.50 \text{ m}} = \frac{2}{3}$$

or $\theta = 41.8°$

Considering the mass,

$$\sum F_y = 0: \quad 2T\cos\theta = mg$$



ANS. FIG. P18.85

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or $$T = \frac{(12.0\text{ kg})(9.80\text{ m/s}^2)}{2\cos 41.8^\circ} = \boxed{78.9\text{ N}}$$

(b) The speed of transverse waves in the string is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{78.9\text{ N}}{0.001\,00\text{ kg/m}}} = 281\text{ m/s}$$

For the standing wave pattern shown (3 loops), $d = \frac{3}{2}\lambda$,

or $$\lambda = \frac{2(2.00\text{ m})}{3} = 1.33\text{ m}.$$

Thus, the required frequency is $$f = \frac{v}{\lambda} = \frac{281\text{ m/s}}{1.33\text{ m}} = \boxed{211\text{ Hz}}.$$

P18.86 (a) Let θ (refer to ANS. FIG. P18.85) represent the angle each slanted rope makes with the vertical. In the diagram, observe that:

$$\sin\theta = \frac{d/2}{(L-d)/2} = \frac{d}{L-d}$$

and

$$\cos\theta = \sqrt{1-\sin^2\theta} = \left[1-\left(\frac{d}{L-d}\right)^2\right]^{\frac{1}{2}}$$

$$\cos\theta = \left[\frac{(L^2-2dL+d^2)-d^2}{(L-d)^2}\right]^{\frac{1}{2}}$$

$$\cos\theta = \frac{\sqrt{L^2-2dL}}{L-d}$$

Considering the mass,

$$\sum F_y = 0:\quad 2T\cos\theta = mg \rightarrow T = \frac{mg}{2\cos\theta}$$

or $$T = \boxed{\frac{mg(L-d)}{2\sqrt{L^2-2dL}}}.$$

(b) The speed of transverse waves in the string is $v = \sqrt{\frac{T}{\mu}}$.

For the standing wave pattern shown (3 loops), $d = \frac{3}{2}\lambda$,

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or $\lambda = \dfrac{2d}{3}$.

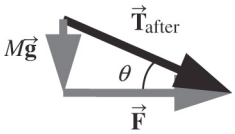
Thus, the required frequency is $f = \dfrac{v}{\lambda} = \boxed{\dfrac{3}{2d}\sqrt{\dfrac{mg(L-d)}{2\mu\sqrt{L^2-2dL}}}}$.

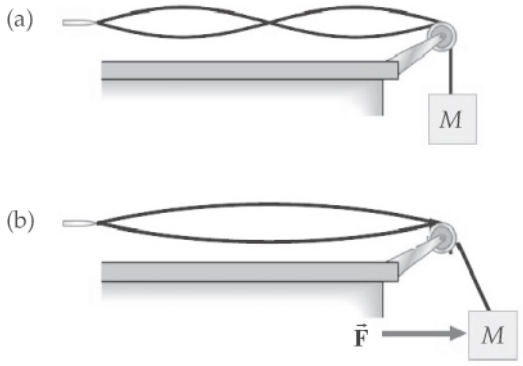
Challenge Problems

P18.87 The idea is that the tension on the string after the force is applied is the vector sum of the wind force $\vec{\mathbf{F}}$ and the weight $M\vec{\mathbf{g}}$ of the mass.

$$\vec{\mathbf{T}}_{\text{after}} = \vec{\mathbf{F}} + M\vec{\mathbf{g}}$$

Notice that this forms a right triangle:





ANS. FIG. P18.87

The relationship between the driving frequency and the string tension is

$$f = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$$

Before and after the application of the wind force, the frequency f, string mass density μ, and string length L are all held constant. Thus, the string tension T is a function of only one variable, n.

$$f = \frac{n_1}{2L}\sqrt{\frac{T_1}{\mu}}, \quad f = \frac{n_2}{2L}\sqrt{\frac{T_2}{\mu}}; \quad \text{thus} \quad 2Lf\sqrt{\mu} = n_1\sqrt{T_1} = n_2\sqrt{T_2}$$

where $n_1 = 2$ and $n_2 = 1$:

$$T_{\text{after}} = T_2 = T_1\left(\frac{n_1}{n_2}\right)^2 = Mg\left(\frac{n_1}{n_2}\right)^2 = Mg\left(\frac{2}{1}\right)^2 = 4Mg$$

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From the geometry of the right triangle,

$$T_2^2 = F^2 + (Mg)^2$$
$$(4Mg)^2 = F^2 + (Mg)^2 \rightarrow F^2 = 16(Mg)^2 - (Mg)^2 = 15(Mg)^2$$
$$F = \boxed{\sqrt{15}Mg}$$

P18.88 Equation 18.13 is

$$y(t) = \sum\left(A_n \sin 2\pi f_n t + B_n \cos 2\pi f_n t\right)$$
$$= \sum\left(A_n \sin n\omega t + B_n \cos n\omega t\right)$$

(a) Multiplying by $\sin m\omega t$ gives:

$$y(t)\sin m\omega t = \sum \sin m\omega t\left(A_n \sin n\omega t + B_n \cos n\omega t\right)$$

Integrating over one period T gives:

$$\int_0^T y(t)\sin m\omega t dt = \sum \int_0^T A_n (\sin n\omega t)(\sin m\omega t) dt + \sum \int_0^T B_n (\cos n\omega t)(\sin m\omega t) dt \qquad \textbf{[1]}$$

Inspecting the left-hand side of the equation, we note that $y(t)$ is a positive constant A for half of the period T, and an equal but negative constant $-A$ for the other half period:

$$\int_0^T y(t)\sin m\omega t dt = \int_0^{T/2} A \sin m\omega t dt + \int_{T/2}^T -A \sin m\omega t dt$$

If we look at the first of the two integrals on the right:

$$\int_0^{T/2} A \sin m\omega t dt = -\frac{A}{m\omega}\cos m\omega t \Big|_0^{T/2}$$
$$= -\frac{A}{m\omega}\left[\cos m\omega\left(\frac{T}{2}\right) - \cos m\omega(0)\right]$$

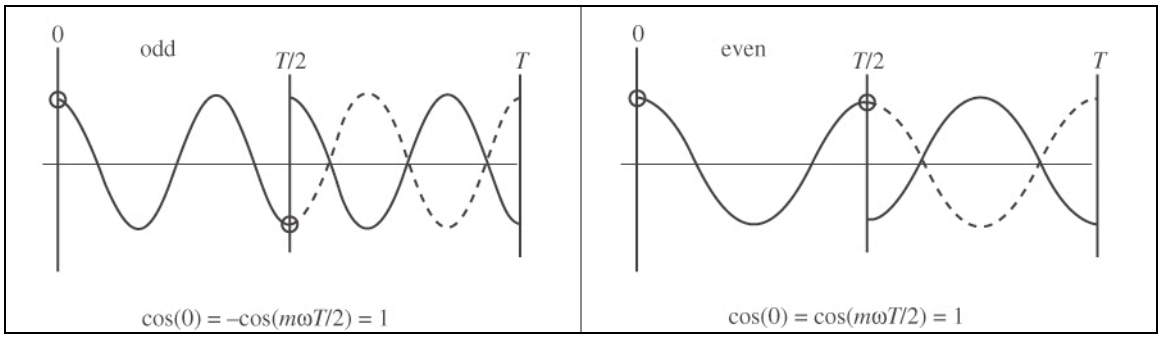
which gives different answers depending on whether m is even or odd:

If m is odd: $= -\frac{A}{m\omega}\left[(-1) - (1)\right] = \boxed{\frac{2A}{m\omega}}$

If m is even: $= -\frac{A}{m\omega}\left[(1) - (1)\right] = \boxed{0}$

(because we are integrating over half periods).

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(a) **(b)**

ANS. FIG. P18.88

The second of the two integrals on the right gives a similar result:

$$\int_{T/2}^{T} -A\sin m\omega t dt = -\left(-\frac{A}{m\omega}\right)\cos m\omega t\Big|_{T/2}^{T}$$

$$= +\frac{A}{m\omega}\left[\cos m\omega(T) - \cos m\omega\left(\frac{T}{2}\right)\right]$$

$$= \begin{cases} \left(\dfrac{2A}{m\omega}\right) & \text{odd} \\ (0) & \text{even} \end{cases}$$

Thus,

$$\int_0^T y(t)\sin m\omega t dt = \int_0^{T/2} A\sin m\omega t dt + \int_{T/2}^{T} -A\sin m\omega t dt$$

$$= \begin{cases} \dfrac{2A}{m\omega} & m \text{ odd} \\ 0 & m \text{ even} \end{cases} + \begin{cases} \dfrac{2A}{m\omega} & m \text{ odd} \\ 0 & m \text{ even} \end{cases}$$

Putting everything together, we have shown that

$$\int_0^T y(t)\sin m\omega t dt = \begin{cases} \dfrac{4A}{m\omega} & m \text{ odd} \\ 0 & m \text{ even} \end{cases}$$

(b) We can analyze the terms involving B_n on the right hand side of eqn. [1] above:

$$\sum \int_0^T B_n(\cos n\omega t)(\sin m\omega t)dt$$

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Using the trigonometric identity

$$\cos\alpha\sin\beta = \frac{1}{2}\sin(\alpha+\beta) - \frac{1}{2}\sin(\alpha-\beta)$$

we have

$$\sum\int_0^T B_n(\cos n\omega t)(\sin m\omega t)\,dt$$

$$= \sum\frac{1}{2}B_n\int_0^T\left[\sin(n\omega t + m\omega t) - \sin(n\omega t - m\omega t)\right]dt$$

$$= \sum\frac{1}{2}B_n\int_0^T\left[\sin(n+m)\omega t - \sin(n-m)\omega t\right]dt$$

The sine function, whether the terms are $(n + m)$ or $(n - m)$, it will always integrate to zero over any full multiple of a period:

$$= \sum\frac{1}{2}B_n\left[\sin(n+m)\omega t - \sin(n-m)\omega t\right]\Big|_0^T = \sum\frac{1}{2}B_n(0) = 0$$

Thus, all the terms involving B_n on the right hand side of eqn. [1] are equal to zero:

$$\sum\int_0^T B_n(\cos n\omega t)(\sin m\omega t)\,dt = 0$$

(c) For all the terms on the right hand side of eqn.(1) with A_n:

$$\sum\int_0^T A_n(\sin n\omega t\ )(\sin m\omega t)\,dt$$

Using the trigonometric identity

$$\sin\alpha\sin\beta = \frac{1}{2}\cos(\alpha-\beta) - \frac{1}{2}\cos(\alpha+\beta)$$

we have

$$\sum\int_0^T A_n(\sin n\omega t\ )(\sin m\omega t)\,dt$$

$$= \sum\frac{1}{2}A_n\int_0^T\left[\cos(n\omega t - m\omega t) - \cos(n\omega t + m\omega t)\right]dt$$

$$= \sum\frac{1}{2}A_n\int_0^T\left[\cos(n-m)\omega t - \cos(n+m)\omega t\right]dt$$

which can be integrated and evaluated at 0 and T:

$$= \sum\frac{1}{2}A_n\left[\frac{1}{(n-m)\omega}\sin(n-m)\,\omega t - \frac{1}{(n+m)\omega}\sin(n+m)\omega t\right]_0^T$$

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The second term, when evaluated at 0 and T, always gives zero. The same is true for the first term for all values of n except where $n = m$. Thus, all the terms on the right hand side of eqn. (1) with A_n are zero except when $m = n$.

(d) For $n = m$, we will do the integration separately:

$$\sum \int_0^T A_n(\sin n\omega t)(\sin m\omega t)dt + \sum \int_0^T B_n(\cos n\omega t)(\sin m\omega t)dt$$

$$= \sum \int_0^T A_n(\sin n\omega t)(\sin m\omega t)dt + 0$$

$$= \frac{1}{2}A_{n=m}\int_0^T [\cos(n-m)\omega t]dt = \frac{1}{2}\left[A_m \int_0^T \cos(0)dt\right]$$

$$= \frac{1}{2}A_m \int_0^T (1)dt = \frac{A_m}{2}[T-0] = \frac{1}{2}A_m T$$

Thus, the entire right side reduces to $\frac{1}{2}A_m T$.

(e) Starting with our original Equation 18.13:

$$y(t) = \sum \left(A_n \sin n\omega t + B_n \cos n\omega t\right)$$

notice that $y(t)$ is an odd function of t: $y(t) = -y(t)$, and the sine function is also odd, but the cosine function is even. From these observations, we can conclude that there are no cosine terms in the Fourier series expansion of $y(t)$; therefore, all the $B_n = 0$. Thus,

$$y(t) = \sum A_n \sin n\omega t$$

But we have shown in part (a) above that:

$$\int_0^T y(t)\sin m\omega t dt = \frac{4A}{m\omega}$$

where m must be odd, and in part (d) that:

$$\int_0^T y(t)\sin m\omega t dt = \sum \int_0^T \sin m\omega t\left(A_n \sin n\omega t + B_n \cos n\omega t\right)dt$$

$$= \frac{1}{2}A_m T$$

where $n = m$.

Thus, for each A_n term: $\frac{1}{2}A_n T = \frac{4A}{n\omega}$. And because $\omega = \frac{2\pi}{T}$,

$$\frac{4A}{n\omega} = \frac{1}{2}A_n T \rightarrow A_n = \frac{8A}{n\omega T} = \frac{4A}{n\pi}\left(\frac{2\pi}{\omega T}\right) = \frac{4A}{n\pi}$$

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which we substitute in to give:

$$\boxed{y(t) = \sum_n \frac{4A}{n\pi} \sin n\omega t}$$

where the summation is only over odd values of n.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P18.2 See ANS. FIG. P18.2.

P18.4 (a) See ANS. FIG. P18.4 (a-e); (b) See ANS. FIG. P18.4 (f-j)

P18.6 (a) $\lambda/4 = 0.113$ m; (b) $\lambda/2 = 0.227$ m

P18.8 (a) 3.33 rad; (b) 283 Hz

P18.10 The man walks only through two minima; a third minimum is impossible.

P18.12 0.500 s

P18.14 (a) The separation of adjacent nodes is $\Delta x = \frac{\pi}{k} = \frac{\lambda}{2}$. The nodes are still separated by half a wavelength; (b) Yes. The nodes are located at $kx + \frac{\phi}{2} = n\pi,$ so that $x = \frac{n\pi}{k} - \frac{\phi}{2k},$ which means that each node is shifted $\frac{\phi}{2k}$ to the left by the phase difference between the traveling waves in comparison to the case in which $\phi = 0.$

P18.16 See P18.16 for full verification.

P18.18 (a) See ANS. FIG. P18.18; (b) In any one picture, the wavelength is the smallest distance along the x axis that contains a nonrepeating shape. The wavelength is $\lambda = 4$ m; (c) The frequency is the inverse of the period. The period is the time the wave takes to go from a full amplitude starting shape to the inversion of that shape and then back to the original shape. The period is the time interval between the top and bottom graphs: 20 ms. The frequency is 1/0.020 s = 50 Hz; (d) 4 m. By comparison with the wave function $y = (2A\sin kx)\cos\omega t$, we identify $k = \pi / 2,$ and then compute $\lambda = 2\pi / k;$ (e) 50 Hz. By comparison with the wave function $y = (2A\sin kx)\cos\omega t$, we identify $\omega = 2\pi f = 100\pi$.

P18.20 (a) 0.600 m; (b) 30.0 Hz

P18.22 (a) 495 Hz; (b) 990 Hz

P18.24 (a) 5.20 m; (b) No. We do not know the speed of waves on the string.

P18.26 (a) 78.6 Hz; (b) 157 Hz, 236 Hz, 314 Hz

P18.28 (a) 4.90×10^{-3} kg/m; (b) 2; (c) no standing wave will form

P18.30 $m = \frac{Mg\cos\theta}{4f^2L}$

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P18.32 291 Hz

P18.34 12 h, 24 min. The natural frequency of the water sloshing in the bay agrees precisely with that of lunar excitation, so we identify the extra-high tides as amplified by resonance.

P18.36 9.00 Hz

P18.38 2.94 cm

P18.40 (a) 536 Hz; (b) 42.9 mm

P18.42 (a) 17.0 Hz; (b) 33.9 Hz; (c) 17.6 Hz, 35.1 Hz

P18.44 0.502 m and 0.837 m

P18.46 n(206 Hz) and n(84.5 Hz)

P18.48 (a) 0.085 8n Hz, with n = 1, 2, 3 . . . ; (b) It is a good rule. A car horn would produce several or many of the closely-spaced resonance frequencies of the air in the tunnel, so it would be great amplified.

P18.50 $\dfrac{\pi r^2 v}{2Rf}$

P18.52 It is impossible because a single column could not produce both frequencies.

P18.54 1.16 m

P18.56 (a) 521 Hz or 525 Hz; (b) 526 Hz; (c) reduced by 1.14%

P18.58 85.7 Hz

P18.60 See P18.60 for a table of the frequencies of the harmonics of each note. The second harmonic of E is close to the third harmonic of A, and the fourth harmonic of C# is close to the fifth harmonic of A.

P18.62 (a) 0.522 m; (b) 316 Hz

P18.64 146 Hz

P18.66 (a) 5.0 Hz, 10.0 Hz, 15.0 Hz; (b) The frequency could be the fifth state at 25.0 Hz or any integer multiple, such as the tenth state at 50.0 Hz, the fifteenth state at 75.0 Hz, and so on.

P18.68 (a) larger; (b) 2.43

P18.70 (a) the particle under constant acceleration model; (b) waves under boundary conditions model; (c) $Mg \sin\theta$; (d) $h\left(\dfrac{1+\sin\theta}{\sin\theta}\right)$; (e) $\dfrac{m\sin\theta}{h(1+\sin\theta)}$; (f) $\sqrt{\dfrac{Mgh}{m}(1+\sin\theta)}$; (g) $\sqrt{\dfrac{Mg}{4mh}(1+\sin\theta)}$; (h) 121 Hz; (i) 60.6 Hz

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P18.72 (a) greatest integer $\leq d\left(\frac{f}{v}\right)+\frac{1}{2}$;

(b) $L_n = \dfrac{d^2 - \left(n-\frac{1}{2}\right)^2\left(\frac{v}{f}\right)^2}{2\left(n-\frac{1}{2}\right)\left(\frac{v}{f}\right)}$, where $n = 1,\ 2,\ \ldots,\ n_{\max}$

P18.74 (a) $\dfrac{d\lambda}{dt} = \dfrac{d}{dt}(2L) = \dfrac{d}{dt}\left[2\left(L_0 + \frac{1}{2}at^2\right)\right] = 2at = 2\left(0.800\ \text{m/s}^2\right)(1.20\ \text{s}) = 1.92$ m/s; (b) 0.960 m/s, half as much as for the first harmonic; (c) Yes. A yo-yo of different mass will hold the string under different tension to make each string wave vibrate with a different frequency, but the geometrical argument given in part (a) still applies to the wavelength; (d) Yes, for the same reason as (c); the geometrical argument given in part (b) still applies to the wavelength.

P18.76 (a) 14.3 m/s; (b) 86.0 cm, 28.7 cm, 17.2 cm; (c) 4.14 Hz, 12.4 Hz, 20.7 Hz

P18.78 (a) 3.99 Hz; (b) 3.99 Hz

P18.80 (a) 59.9 Hz; (b) 20.0 cm

P18.82 (a) frequency should be halved; (b) $\left[\dfrac{n}{n+1}\right]^2 T$; (c) $\dfrac{T'}{T} = \dfrac{9}{16}$

P18.84 (a) 0.656 m; (b) 13.5° C

P18.86 (a) $\dfrac{mg(L-d)}{2\sqrt{L^2-2dL}}$; (b) $\dfrac{3}{2d}\sqrt{\dfrac{mg(L-d)}{2\mu\sqrt{L^2-2dL}}}$

P18.88 (a) see P18.88(a) for full explanation; (b) see P18.88(b) for full explanation; (c) See P18.88(c) for full explanation; (d) see P18.88(d) for full explanation; (e) see P18.88(e) for full explanation.

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